

Nearly Optimal Community Recovery by Spectral Likelihood Decoding

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Abstract

Spectral clustering compresses a network into a small number of eigenpairs, whereas likelihood-ratio classification uses block-dependent logarithmic contrasts. We show that the leading Chernoff error exponent can be attained by a decoder that uses only the retained adjacency eigenpairs. Our procedure fits a Poisson-form working mixture to positive reconstructed connection profiles. A growing initialization starts at the global profile mean, adds candidate profiles by their likelihood gain on all nodes, and refits after each addition. For a stochastic block model in which $(n/\log n)P$ is fixed, entrywise positive, symmetric and full rank, with positive limiting community proportions and an admissible fixed positive profile floor, we prove end-to-end recovery for this growing procedure when $K = 2$. For general fixed K , we add an explicit safeguard: $\lceil \log n \rceil$ independent likelihood-residual++ initializations, fitted only after all K seeds have been selected, compete with the growing fit under the same observed working mixture likelihood. One final EM update and likelihood-ratio decoding give $\mathbb{E} \text{Mis}(\hat{z}, z) \leq e^{-(1+o(1))I_n}$ without an assumed weak initializer or a K -means warm start. The guarantee concerns the leading exponent, with I_n the minimum pairwise Chernoff information. The proof combines empirical spectral linearization, likelihood-based initialization and fit selection, and invariant-basin EM analysis for the dependent spectral profiles. Experiments on two paired collections totaling 240 graphs evaluate the original growing and parameter-replacement procedures; the safeguarded combination has not been evaluated on that full collection. We also state the additional requirements for an extension below logarithmic density.

1 Introduction

Community recovery in the stochastic block model (SBM) is a basic problem in network statistics. The model assigns each node to an unknown community and makes edge probabilities depend on the communities of their endpoints [6]. Two common approaches use different representations of the same data. Spectral methods reduce the adjacency matrix to a small number of eigenvectors and then cluster the embedded nodes. Likelihood methods compare how well alternative community assignments explain the observed connections. We show that likelihood-ratio information can be decoded from the retained spectrum to achieve nearly optimal community recovery in a general fixed- K SBM at logarithmic density, without returning to the original adjacency rows.

This question concerns the decoding rule as well as the embedding. A low-rank approximation is selected by a quadratic criterion, and standard spectral clustering applies a further Euclidean criterion through K -means. The oracle classifier for an SBM instead weights connections by logarithms of block probabilities and includes the contribution of absent edges. For a general

connectivity matrix and unequal community sizes, its decision boundaries need not agree with nearest-center boundaries in the spectral embedding. Nevertheless, the fact that spectral reduction uses a quadratic objective does not itself imply that it discards the relevant likelihood information: a projection preserves every linear statistic whose weight vector belongs to the projected subspace.

Optimal recovery and likelihood refinement. The information-theoretic limits of the SBM are well developed. Abbe and Sandon [1] characterize exact recovery in the general logarithmic-density SBM through Chernoff–Hellinger divergence and give efficient algorithms reaching that threshold. Zhou and Li [13] study the optimal error rate through Chernoff bounds and its application to community detection. We use these works to identify the oracle benchmark, while the statistical objective of the present paper is the leading-exponent form

$$\exp\{-(1 + o(1))I\},$$

where I is the relevant pairwise Chernoff information. Throughout this paper, “nearly optimal” means matching this leading exponent.

Spectral initialization followed by likelihood refinement is also an established approach. Amini, Chen, Bickel and Levina [3] fit mixtures to blockwise connection counts using pseudo-likelihood and iterative updates. Gao, Ma, Zhang and Zhou [5] prove that a weakly consistent initializer followed by penalized local likelihood refinement achieves optimal misclassification performance under their assumptions. Zhou and Amini [12] obtain optimal bipartite recovery using spectral initialization and pseudo-likelihood classification, and analyze a broader family of EM-type biclustering updates. In these methods the refinement uses information calculated from the original edge observations. Our algorithm imposes an additional restriction: once the selected adjacency eigenpairs have been computed, initialization, parameter updates, classification and component replacement use only those eigenpairs and quantities reconstructed from them.

Likelihood information in spectral representations. Spectral optimality is known in important special cases. Abbe, Fan, Wang and Zhong [2] establish a first-order entrywise approximation for eigenvectors and show that, in the equal-sized symmetric two-community SBM at logarithmic density, an untrimmed spectral method attains the optimal recovery threshold and misclassification exponent. Their result already demonstrates that a spectral representation can retain the decisive local evidence. For sparse homogeneous SBMs with multiple, possibly unequal communities, Zhang [10] gives matching upper and lower error exponents for adjacency spectral clustering with K -means. His spectral exponent is distinguished from the information-theoretic exponent, so a sharp analysis of a spectral algorithm does not automatically establish its statistical optimality.

Probabilistic inference based on spectral embeddings predates the present work. Suwan et al. [9] use Gaussian-mixture limits of adjacency spectral embedding to construct an empirical Bayes prior for block membership inference. This is a relevant example of using distributional information beyond Euclidean clustering; it does not establish a sparse large-deviation equivalence between our spectral scores and the oracle edge likelihood ratio. For lower densities, Zhou and Amini [11] provide data-driven adjacency regularization and consistency analysis in the general bipartite setting. These results supply useful tools for the extension considered in Section 4, but regularized spectral concentration by itself does not prove preservation of the oracle error exponent.

Our approach. We retain the K eigenpairs of the adjacency matrix with largest absolute eigenvalues, including signal eigenvalues of either sign, and form positive reconstructed connection

profiles q_i . A Poisson-form working score compares these profiles with candidate community profiles. Its pairwise difference has the algebraic form of a likelihood ratio. Our growing initialization starts a single component at the global profile mean, adds a node-profile candidate according to its gain over all nodes, and refits every component after each addition. This criterion makes repeated representation of an already well-explained region compete against improvement in omitted regions.

For two communities, we prove that this growing procedure itself attains the leading oracle exponent (Theorem 3.8). For general fixed K , Algorithm 1 includes a stated initialization safeguard. Alongside the growing fit, it fits $\lceil \log n \rceil$ independent likelihood-residual++ initializations. Each draws its first node uniformly and draws each subsequent seed with probability proportional to

$$\min_{c \in S} D_{\text{Pois}}(q_i \| q_c),$$

where S is the set of seeds already selected. Each such initialization selects all K seeds before running EM. We select the fitted candidate with greatest observed working mixture likelihood, then perform one final EM update and decode by likelihood ratios. This is an explicit addition to the growing procedure; the likelihood criteria, parameter updates, and final selection continue to use only the retained eigenpairs.

The interpretation as a spectral likelihood ratio has a precise algebraic basis. If $\hat{\Pi}$ denotes the projector onto the retained empirical eigenspace, its signed rank- K reconstruction equals $A\hat{\Pi}$. With a population SBM projector, every block-constant likelihood-ratio contrast is preserved when the block matrix is full rank. Empirical projection, positivity clipping and estimated parameters introduce additional errors. We control those errors for the whole-graph empirical profiles through first-order entrywise eigenspace analysis and deterministic likelihood and EM bounds. The reconstructed profiles are dependent, noninteger quantities, so the numerical fitting objective remains a working likelihood. Its relation to the oracle Bernoulli likelihood is established by the statistical analysis.

The paper makes three contributions. First, it gives a spectral-only likelihood decoder with a complete initialization rule. The growing rule is justified directly for $K = 2$; for general fixed K , the growing fit competes with the independent likelihood-residual++ starts under a common fitting objective. This procedure does not require a K -means warm start. Second, Theorem 3.9 proves the end-to-end bound

$$\mathbb{E} \text{Mis}(\hat{z}, z) \leq \exp\{-(1 + o(1))I_n\}$$

under the stated fixed full-rank SBM assumptions, logarithmic density, and an admissible fixed positive profile floor. Here I_n is the minimum pairwise oracle Chernoff information. The theorem does not assume that initialization has already reached a weak-recovery basin: the proof establishes the needed initialization and likelihood-selection properties and then applies the local invariant-basin result. It implies exact recovery when the normalized minimum information exceeds the logarithmic threshold. The local EM theorem and algebraic preservation results serve as tools for this full recovery theorem, rather than substitutes for an initialization guarantee.

Third, paired experiments on 240 graphs evaluate the original growing rule and likelihood-accepted parameter replacement. They diagnose severe local optima in which nearby communities share one mixed profile while another community receives two components, and show how all-node likelihood gains or accepted replacements repair the observed failures. These results concern those evaluated procedures. They are not presented as a full empirical evaluation of the newly safeguarded Algorithm 1. Establishing an equally general theorem for the growing rule alone, without the safeguard, remains an extension of the main result.

Section 2 specifies the model, spectral scores and algorithms. Section 3 gives the main results without adjacency regularization in the logarithmic-density regime. Section 4 formulates the lower-

density extension and its additional requirements. Section 5 presents experiments and failure diagnostics, and Section 6 concludes. Proofs appear in the appendix.

2 Spectral likelihood-ratio decoding

2.1 Model and retained information

Let A be the adjacency matrix of an undirected graph without self-loops. Conditional on community labels $z = (z_1, \dots, z_n) \in [K]^n$, its upper-triangular entries are independent and

$$A_{ij} \sim \text{Bernoulli}(P_{z_i z_j}), \quad i < j, \quad A_{ji} = A_{ij}, \quad A_{ii} = 0. \quad (2.1)$$

We follow the SBM notation of Zhou and Li [13]: $P \in (0, 1)^{K \times K}$ is the block probability matrix, $p_{kj} = P_{k, z_j}$, and $p_{k*} = (p_{k1}, \dots, p_{kn})$ is the connection profile of community k . The number $K \geq 2$ is fixed and known. Write $Z_{ia} = \mathbf{1}\{z_i = a\}$, $n_a = \sum_i Z_{ia}$, and $\pi_{a,n} = n_a/n \rightarrow \pi_a > 0$. The matrix P may depend on n . In the logarithmic-density regime, put $d_n = \log n$ and assume that $(n/d_n)P$ is fixed, symmetric and full rank, and define

$$b_- = \min_{a,b} \frac{nP_{ab}}{d_n} > 0, \quad b_+ = \max_{a,b} \frac{nP_{ab}}{d_n} < \infty.$$

Full rank need not imply positive definiteness; disassortative signal directions are retained.

The $n \times n$ mean matrix differs from the rank- K matrix $ZPZ^\top$ only on its diagonal:

$$\mathbb{E}[A \mid z] = ZPZ^\top - \text{diag}(P_{z_i z_i}). \quad (2.2)$$

The small diagonal correction matters for exact identities. The representation consists of the K empirical eigenpairs of A with largest absolute eigenvalues:

$$AU = U\Lambda, \quad U^\top U = I_K, \quad \tilde{A} = U\Lambda U^\top. \quad (2.3)$$

All subsequent fitting accesses only (U, Λ) . This restriction excludes an additional edge-based refinement. It also excludes computing separate leave-one-out eigendecompositions as an undeclared algorithmic step. In the logarithmic regime we use the adjacency eigenpairs without degree regularization or separately stored degrees.

2.2 Positive profiles and a working likelihood

We construct the positive profiles

$$q_{ij} = \max\{\tilde{A}_{ij}, \varepsilon_n\}, \quad \varepsilon_n = c_0 d_n / n, \quad c_0 = 10^{-4}. \quad (2.4)$$

The evaluated implementation retains the reconstructed diagonal. The logarithmic-density local recovery results assume $c_0 < b_-$; the raw oracle results do not need this additional floor margin. For $\hat{p} \in (0, \infty)^n$ define

$$\ell_i(\hat{p}) = \sum_{j=1}^n q_{ij} \log \hat{p}_j - \sum_{j=1}^n \hat{p}_j. \quad (2.5)$$

Observation-only terms, including Gamma constants, have been omitted. They cancel from the all-node gains and responsibility comparisons and do not change the mixture maximizer. Although this

is the algebraic form of a Poisson log likelihood, q_{ij} are dependent, generally noninteger quantities. We therefore use it as a *working* score, rather than assert that the spectral observations follow an independent Poisson model.

For profiles $\hat{p}_1, \dots, \hat{p}_K$ and weights $\hat{\pi} \in \Delta_K$, the working mixture objective is

$$\mathcal{L}(\hat{\pi}, \hat{p}) = \sum_{i=1}^n \log \left(\sum_{a=1}^K \hat{\pi}_a e^{\ell_i(\hat{p}_a)} \right). \quad (2.6)$$

The final label is

$$\hat{z}_i = \arg \max_{a \in [K]} \{ \ell_i(\hat{p}_a) + \log \hat{\pi}_a \}. \quad (2.7)$$

Thus classification uses pairwise working likelihood ratios, with estimated prior weights determining the thresholds. We evaluate the permutation-invariant mislabeled fraction

$$\text{Mis}(\hat{z}, z) = \min_{\sigma \in \mathfrak{S}_K} \frac{1}{n} \sum_{i=1}^n \mathbf{1}\{\sigma(\hat{z}_i) \neq z_i\}, \quad (2.8)$$

where $\mathfrak{S}_K$ is the set of label permutations. The fitted profiles $\hat{p}_{a*}$ are responsibility-weighted means of the observed profiles; the implementation does not impose the exact SBM constraint $\hat{p}_{aj} = P_{a,z_j}$. Although these profiles have n coordinates, every M-step keeps them in the row convex hull of q ; Proposition B.4 records this restriction.

2.3 Global-mean start and growing likelihood gains

The procedure starts with the global profile mean,

$$\hat{p}_1^{(1)} = \frac{1}{n} \sum_{i=1}^n q_i, \quad \hat{\pi}_1^{(1)} = 1. \quad (2.9)$$

This is the fitted one-component solution, rather than a node chosen using a top-fraction subset. It requires no subset-size parameter and removes no observations.

At stage $t = 2, \dots, K$, let $\hat{p}_1^{(t-1)}, \dots, \hat{p}_{t-1}^{(t-1)}$ be the profiles fitted at the preceding stage and set $h_i^{(t-1)} = \max_{a < t} \ell_i(\hat{p}_a^{(t-1)})$. For a node candidate c not previously added, define

$$\Delta_t(c) = \sum_{i=1}^n (\ell_i(q_c) - h_i^{(t-1)})_+, \quad c_t \in \arg \max_c \Delta_t(c). \quad (2.10)$$

Append q_{c_t} to the current $t - 1$ fitted profiles, initialize the t weights uniformly, and fit all t profiles jointly on all n observations by the updates below. The fitted endpoint supplies the next stage's profiles. A global mean counts as the first component, so only $K - 1$ node candidates are added. Previously chosen node IDs are excluded from subsequent candidate scans, and ties are broken deterministically.

The all-node gain rewards a candidate for explaining evidence not already well fitted. This is a deterministic analogue of the coverage principle behind *k-means++* [4], with likelihood gains replacing squared Euclidean distances. It does not guarantee that a selected node is a population community center.

For fixed profiles, Proposition B.1 gives a submodular greedy bound and Proposition B.2 gives noiseless coverage from a node-profile first seed. They describe the frozen-profile control, not the

complete growing algorithm: the latter starts from a mean and changes its profiles between additions. The original empirical primary branch is the growing procedure specified here. It supplies one candidate fit for the complete decoder below; the additional likelihood restarts make the general- K guarantee independent of its basin dynamics.

2.4 Likelihood-residual restarts and the complete decoder

For general K , we supplement the growing fit with $L_n = \lceil \log n \rceil$ independent likelihood-residual initializations. Define the Poisson deviance

$$D_P(x||y) = \sum_j \{x_j \log(x_j/y_j) - x_j + y_j\}.$$

Each restart chooses its first node uniformly. Given a set C of selected nodes, it chooses the next node with probability proportional to

$$r_i(C) = \min_{c \in C} D_P(q_i||q_c).$$

Already selected nodes have zero sampling mass. If all residuals vanish, choose an unselected node uniformly. Select all K seeds before fitting; initialize the profiles at their q rows and weights at $1/K$, and run the same exact joint EM updates as for the growing fit. The random streams are independent conditional on q .

Compare the endpoints of these L_n fits and the original growing fit using exactly the same objective $\mathcal{L}$ in (2.6). Keep the largest value, compute its E-step responsibilities, perform one additional profile and weight M-step, and decode. This final update is part of the algorithm, even when an earlier fit stopped at its iteration cap. Further exact updates are optional. All profiles remain convex combinations of the same positive q rows; a zero-mass component retains its previous profile.

This is a likelihood-based restart safeguard, not a Euclidean warm start. It is an explicit addition to the original growing branch. Theorem 3.9 proves the complete general- K algorithm without an initialization assumption. For $K = 2$, Theorem 3.8 shows that the growing branch alone already has the target guarantee.

2.5 Alternating fitting and parameter replacement

At a stage with s components, initialize $\hat{\pi}_a = 1/s$ and perform soft updates (replace K by s in the displayed sums)

$$\tau_{ia} = \frac{\hat{\pi}_a e^{\ell_i(\hat{p}_a)}}{\sum_b \hat{\pi}_b e^{\ell_i(\hat{p}_b)}}, \quad N_a = \sum_i \tau_{ia}, \quad (2.11)$$

$$\hat{p}_{aj}^{\text{new}} = \frac{\sum_i \tau_{ia} q_{ij}}{N_a}, \quad \hat{\pi}_a^{\text{new}} = N_a/n. \quad (2.12)$$

The responsibilities are computed using log-sum-exp for numerical stability. Positive q ensures that positive-weight components have positive updated means. The evaluated implementation constrains $\hat{\pi}_a \geq 10^{-8}$ and replaces the unconstrained weight update in (2.12) by the exact optimizer of $\sum_a N_a \log \hat{\pi}_a$ on that floored simplex. If no floor is active, this reduces to N_a/n . Both exact versions maximize the variational M-step, giving objective monotonicity and boundedness within

that stage. Proposition B.3 proves convergence of the objective within each fit; it does not prove global optimality or correct community recovery. Because the implementation reinitializes weights when it adds a component, we do not assert monotonicity between stages.

Algorithm 1: Spectral likelihood decoding with a restart safeguard

1. Input K and the retained eigenpairs (U, Λ) ; form q by (2.4).
2. Obtain the global-mean growing fit by (2.9), (2.10), and the joint EM updates.
3. Independently run $L_n = \lceil \log n \rceil$ likelihood-residual initializations as in Section 2.4; select all K seeds before each joint fit.
4. Select the largest final $\mathcal{L}$ across the growing fit and all restarts.
5. Compute one E-step and one profile and weight M-step on the selected fit. Decode by (2.7), or continue exact EM before decoding.
6. Output labels, fitted profiles, weights, and the recorded objectives.

We also evaluate an alternative stabilization of the earlier partial-peeling initialization. After fitting, remove each component in turn; choose a replacement profile by its global gain relative to the remaining fitted profiles, and run a fresh fit. Accept the new fit only if its final working mixture objective increases. The evaluated replacement pass tests the K possible component removals and selects an improving trial. This is an alternative branch with additional fitting cost; the results below do not assert that the combination of Algorithm 1 and replacement was evaluated.

The numerical experiments use the exact floored-simplex weight update, finite stopping tolerances, and an iteration cap. Objective convergence in exact arithmetic does not replace checking the finite-run stopping rule or floating-point error. We report those numerical checks separately.

2.6 Why the representation can retain an LLR

Suppose temporarily that P and the other labels are known. For testing $z_i = a$ against $z_i = b$, define, for $j \neq i$,

$$w_j^{ab} = \log \frac{p_{aj}(1 - p_{bj})}{p_{bj}(1 - p_{aj})}, \quad c_i^{ab} = \sum_{j \neq i} \log \frac{1 - p_{aj}}{1 - p_{bj}}. \quad (2.13)$$

The oracle Bernoulli LLR is $L_i^{ab} = \sum_{j \neq i} A_{ij} w_j^{ab} + c_i^{ab}$. Extend w_i^{ab} by the same block formula. Because $A_{ii} = 0$, this extension does not change the oracle linear statistic. The contrast vector is block constant, hence belongs to $\text{col}(Z)$.

The fundamental identity is

$$\tilde{A} = A\Pi, \quad \Pi = UU^\top, \quad (\tilde{A}_i - A_i)w^{ab} = A_i(\Pi - \text{Id})w^{ab}. \quad (2.14)$$

Here Id denotes the identity matrix. If w^{ab} lies in the retained empirical space, the linear statistic is preserved exactly. At the population level, full rank gives $\text{col}(ZPZ^\top) = \text{col}(Z)$, so every contrast has the required span membership. The spectral decomposition need not itself optimize a likelihood: it retains a linear space containing the directions in which the likelihood acts.

The executable score additionally involves clipping, estimated profiles, mixture thresholds, and a retained reconstructed diagonal. These perturbations require statistical control. A matrix reconstruction bound alone does not establish the oracle exponent; the local fitting theorem uses the stronger row-profile approximation, the clipping moment bound, and a weak-initialization condition. Theorem 3.6 supplies the fitting guarantee. Theorem 3.8 establishes its initialization condition for the original growing branch when $K = 2$, and Theorem 3.9 establishes it for the complete Algorithm 1 for every fixed K .

2.7 Computational cost

Forming the full positive reconstruction stores $O(n^2)$ numbers. Caching all candidate scores uses an n -by- n matrix and, by direct multiplication, $O(n^3)$ arithmetic. Each subsequent greedy scan costs $O(n^2)$, and one EM iteration costs $O(Kn^2)$. Algorithm 1 runs $O(\log n)$ additional fits, so its total remains polynomial for a prescribed polynomial iteration bound. These costs are additional to computing the eigenpairs. The implementation is practical at the tested $n \leq 2000$ but does not yet justify a large-network speed claim. Candidate subsampling and blocked score evaluation are natural approximations, whose statistical and computational performance require separate validation.

3 Main results in logarithmic density

This section gives the exact contrast-preservation identity, the oracle benchmark, an abstract exponent-transfer theorem, and a local recovery theorem for the actual empirical spectral EM fit. It then establishes end-to-end recovery for the original two-community growing branch and for the complete general- K likelihood-restart decoder. A separate corollary records a spectral warm-start alternative.

Throughout this section, K is fixed, $n_a/n \rightarrow \pi_a > 0$, and $d_n \rightarrow \infty$ with $d_n/n \rightarrow 0$. Unless stated otherwise, $(n/d_n)P$ is a fixed symmetric nonsingular matrix whose entries lie in a fixed compact subset of $(0, \infty)$. The logarithmic-density regime is $d_n = \log n$. The probabilistic oracle calculations do not require nonsingularity, but require distinct block rows when all communities are to be classified.

3.1 Spectral information and likelihood contrasts

For communities $a \neq b$, define the block-constant Bernoulli contrast and the nonedge offset by

$$w_j^{ab} = \log \frac{p_{aj}(1 - p_{bj})}{p_{bj}(1 - p_{aj})}, \quad (3.1)$$

$$c_i^{ab} = \sum_{j \neq i} \log \frac{1 - p_{aj}}{1 - p_{bj}}. \quad (3.2)$$

The oracle incident-edge log-likelihood ratio is

$$L_i^{ab} = A_i w^{ab} + c_i^{ab}, \quad (3.3)$$

where A_i is row i of A . Its diagonal coordinate contributes zero.

Proposition 3.1 (Exact preservation of retained contrasts). *Let U contain any orthonormal collection of eigenvectors of the symmetric matrix A , let Λ contain their eigenvalues, and set $\Pi = UU^\top$ and $\tilde{A} = U\Lambda U^\top$. Then $\tilde{A} = A\Pi$ and, for every vector w ,*

$$\tilde{A}_i w - A_i w = A_i(\Pi - \text{Id})w. \quad (3.4)$$

In particular, $w \in \text{col}(U)$ implies exact preservation. Moreover, if P is nonsingular, then the rank- K comparison matrix $ZPZ^\top$ has column space $\text{col}(Z)$, and every w^{ab} belongs to this population signal space.

The proposition explains the likelihood interpretation without requiring the eigendecomposition itself to compute $\log P$. The eigendecomposition retains directions; the likelihood decoder assigns the appropriate contrast coefficients to those directions. It also states the precise limitation: the empirical retained space need not equal the population space. The actual no-self-loop mean is $\mathbb{E}[A \mid z] = ZPZ^\top - \text{diag}(P_{z_i z_i})$; the rank- K statement concerns $ZPZ^\top$, not the no-self-loop mean.

The oracle Poisson contrast is $v_j^{ab} = \log(p_{aj}/p_{bj})$, with score

$$L_{i,P}^{ab} = \sum_{j \neq i} A_{ij} v_j^{ab} - \sum_{j \neq i} (P_{a,z_j} - P_{b,z_j}). \quad (3.5)$$

The following result makes its use in a sparse working score precise.

Lemma 3.2 (Sparse Bernoulli-to-Poisson correction). *There is a constant C depending only on the fixed normalized block probabilities such that, for all sufficiently large n , every i , and every $a \neq b$,*

$$|L_i^{ab} - L_{i,P}^{ab}| \leq C\{(d_n/n) \deg(i) + d_n^2/n\}. \quad (3.6)$$

For every fixed $H > 0$ there is a constant D_H such that $\Pr\{\deg(i) > D_H d_n\} \leq e^{-H d_n}$ uniformly in i . Consequently, the correction is $o(d_n)$ outside an event of probability $e^{-H d_n}$. In logarithmic density its bound on that event is $O(\log^2 n/n) = o(1)$.

3.2 The oracle benchmark

We use Zhou–Li’s profile and Chernoff notation. For $\alpha \in [0, 1]$, define

$$D_\alpha(p_{a,-i} \| p_{b,-i}) = - \sum_{j \neq i} \log \left[p_{aj}^{1-\alpha} p_{bj}^\alpha + (1 - p_{aj})^{1-\alpha} (1 - p_{bj})^\alpha \right], \quad (3.7)$$

$$\alpha_{i,ab}^* \in \arg \max_{\alpha \in [0,1]} D_\alpha(p_{a,-i} \| p_{b,-i}), \quad D_{i,ab}^* = D_{\alpha_{i,ab}^*}(p_{a,-i} \| p_{b,-i}), \quad (3.8)$$

$$I_n = \min_{i, a \neq b} D_{i,ab}^*. \quad (3.9)$$

Here $p_{a,-i}$ omits the absent self-loop coordinate. The notation I_n is our shorthand for the minimum exact pairwise information; it is not a second definition of a rate.

For distinct block rows under the fixed normalized probability model, $I_n \asymp d_n$. In logarithmic density, $I_n / \log n$ converges to a positive finite limit. Zhou and Li [13] studied optimal error rates for SBM community detection. We use their notation and the leading information scale, without restating their refined rate.

Lemma 3.3 (Leading oracle error exponent). *Consider the binary experiment in which $z_i \in \{a, b\}$ has equal prior probabilities and all other labels and P are known. Its Bayes risk $R_{i,ab}^{\text{or}}$ satisfies, uniformly in i ,*

$$R_{i,ab}^{\text{or}} = \exp\{-(1 + o(1))D_{i,ab}^*\}. \quad (3.10)$$

The same leading exponent holds for fixed prior probabilities bounded away from zero and one.

3.3 Transfer to estimated spectral scores

The theorem below accommodates scores estimated from the entire graph, so it does not assume independence of a node’s score and its incident edges. It makes the required rare-event control explicit.

Theorem 3.4 (Likelihood-ratio exponent transfer). *Let $\hat{\ell}_i(a)$ be estimated community scores, after one common alignment of their labels with the population labels, and classify by $\hat{z}_i \in \arg \max_a \hat{\ell}_i(a)$. Suppose there are $\tau_n = o(d_n)$ and $\delta_n \geq 0$ such that, for every i , an event G_i satisfies*

$$\Pr(G_i^c) \leq \delta_n, \quad (3.11)$$

$$\max_{a \neq b} |\hat{\ell}_i(a) - \hat{\ell}_i(b) - L_i^{ab}| \leq \tau_n \quad \text{on } G_i, \quad (3.12)$$

$$\liminf_{n \rightarrow \infty} \frac{-\log \delta_n}{I_n} \geq 1, \quad (3.13)$$

where $-\log 0 = +\infty$. Then the permutation-invariant error fraction obeys

$$\mathbb{E} \text{Mis}(\hat{z}, z) \leq \exp\{-(1 + o(1))I_n\}. \quad (3.14)$$

When $d_n = \log n$ and $\liminf_{n \rightarrow \infty} I_n / \log n > 1$, this implies $\Pr\{\text{Mis}(\hat{z}, z) > 0\} \rightarrow 0$.

The condition on δ_n is essential. An error approximation that holds with probability $1 - o(1)$ can still fail much more frequently than the oracle makes an error, and therefore need not preserve its exponent. The theorem asserts an upper bound on the recovery error. In the binary oracle experiment, data processing and Lemma 3.3 also give the matching leading lower bound for any decoder using only the graph's spectral summary.

For the working scores actually fitted in Section 2, set $\hat{w}_j^{ab} = \log(\hat{p}_{aj}/\hat{p}_{bj})$ and $\hat{c}^{ab} = -\sum_j (\hat{p}_{aj} - \hat{p}_{bj}) + \log(\hat{\pi}_a/\hat{\pi}_b)$. Writing $q_i = \tilde{A}_i + (q_i - \tilde{A}_i)$ yields the exact decomposition

$$\begin{aligned} \hat{\ell}_i(a) - \hat{\ell}_i(b) - L_i^{ab} &= A_i(\Pi - \text{Id})w^{ab} \\ &\quad + \tilde{A}_i(\hat{w}^{ab} - w^{ab}) \\ &\quad + (q_i - \tilde{A}_i)\hat{w}^{ab} + (\hat{c}^{ab} - c_i^{ab}). \end{aligned} \quad (3.15)$$

Thus projection, estimated contrasts, clipping, and offsets must all be controlled at the scale in Theorem 3.4. The sum includes the reconstructed diagonal used by the implemented working score; its discrepancy with the oracle's excluded diagonal is included in the displayed decomposition. Neither average reconstruction accuracy nor objective convergence removes these terms.

3.4 A local recovery theorem for the empirical spectral fit

The next results remove the empirical-projection and parameter-fitting conditions from the preceding abstract transfer statement in logarithmic density. This local statement temporarily assumes reliable initial responsibilities. The end-to-end results below establish that assumption for the original two-community growing branch and for the complete Algorithm 1.

Write $N = Z^\top Z$ and $\Pi_* = ZN^{-1}Z^\top$. For a fixed $c_0 > 0$ satisfying $c_0 < \min_{a,b} nP_{ab}/d_n$, let

$$q^0 = \max\{A\Pi_*, c_0 \frac{d_n}{n}\}, \quad q = \max\{U\Lambda U^\top, c_0 \frac{d_n}{n}\},$$

where the maximum is entrywise. The matrix q^0 is an analysis comparison using the true memberships; it is not an input to the algorithm.

Lemma 3.5 (Full-graph spectral approximation). *Under the fixed normalized full-rank probability model with $d_n = \log n$, for every fixed $H > 0$ there are a constant C_H and a deterministic sequence $a_n \rightarrow 0$ such that, with probability at least $1 - O(n^{-H})$,*

$$\max_i \|q_i - q_i^0\|_1 \leq a_n d_n, \quad c_0 \frac{d_n}{n} \leq q_{ij} \leq C_H \frac{d_n}{n}, \quad (3.16)$$

$$\frac{1}{nd_n} \sum_i \|q_i - p_{z_i,*}\|_1 \leq a_n. \quad (3.17)$$

The reconstruction uses the empirical eigenpairs of the whole graph. No row-independent replacement or extra leave-one-out computation is required by the algorithm.

The first-order entrywise eigenspace theorem of Abbe et al. [2], together with the arbitrary polynomial-confidence spectral norm bound of Lei and Rinaldo [7], supplies the spectral inputs. The appendix verifies their assumptions for the present general full-rank SBM and derives the displayed row-profile bounds.

Theorem 3.6 (Local empirical spectral EM attains the leading exponent). *Use the model and admissible fixed floor of Lemma 3.5. Let the initial responsibilities $R^{(0)} = (R_{ia}^{(0)})$ have nonnegative rows summing to one. They may depend on the entire graph. Suppose that, after a common label alignment, for deterministic $\gamma_n \rightarrow 0$ and $\delta_n \geq 0$,*

$$\Pr \left\{ \frac{1}{n} \sum_{i,a} |R_{ia}^{(0)} - Z_{ia}| > \gamma_n \right\} \leq \delta_n, \quad \liminf \frac{-\log \delta_n}{I_n} \geq 1. \quad (3.18)$$

Perform at least one responsibility-weighted profile M-step on q , followed by the score classification in (2.7). Then

$$\mathbb{E} \text{Mis}(\hat{z}, z) \leq e^{-(1+o(1))I_n}. \quad (3.19)$$

The same bound holds after any further exact EM updates and any finite, possibly data-dependent stopping time after the first M-step. The weights are updated either unconstrained or by the exact fixed floored-simplex optimizer. If a component has zero responsibility mass, define its profile by any fixed positive fallback and use positive weights; on the weak-start event this case does not occur for large n . When $\liminf_{n \rightarrow \infty} I_n / \log n > 1$, the output exactly recovers all labels with probability tending to one.

This theorem proves the statistical behavior of the actual full-graph spectral fitting stage from a sufficiently reliable weak start. It handles positive clipping and the freely fitted profiles rather than assuming that their LLR errors are already small. An initialization proof must establish (3.18): average weak consistency with an unspecified $1 - o(1)$ probability is insufficient. Theorems 3.8 and 3.9 provide the required reliability.

Corollary 3.7 (A fully provable spectral warm-start variant). *Replace the growing initialization by a fixed-factor approximate K -means clustering of the rows of U , computed with failure probability $O(n^{-H})$ for any prescribed fixed $H > \limsup_{n \rightarrow \infty} I_n / \log n$. Convert its labels to hard responsibilities and apply the profile EM decoder. This variant uses only the retained eigenpairs and satisfies (3.19). One polynomial-time implementation uses $O(\log n)$ independent K -means++ seedings and chooses the least squared-error solution.*

This corollary is an alternative initialization. The main end-to-end result below uses likelihood residuals instead. The archived ten-start Lloyd baseline is distinct from either logarithmically amplified initialization.

3.5 End-to-end recovery by likelihood initialization

The following results remove the weak-start assumption from the algorithmic guarantee.

Theorem 3.8 (Original growing decoder for two communities). *Under the model and admissible floor of Lemma 3.5, suppose $K = 2$. The original global-mean growing branch of Section 2.3, followed by at least one exact profile and weight M -step, satisfies*

$$\mathbb{E} \text{Mis}(\hat{z}, z) \leq e^{-(1+o(1))I_n}.$$

This includes its evaluated finite stopping rule, since the first M -step is mandatory. When $\liminf_{n \rightarrow \infty} I_n / \log n > 1$, it exactly recovers all labels with probability tending to one. No likelihood restarts or pure-candidate assumption are needed.

Theorem 3.9 (Complete likelihood decoder attains the leading exponent). *Under the same model for every fixed $K \geq 2$, Algorithm 1 satisfies*

$$\mathbb{E} \text{Mis}(\hat{z}, z) \leq e^{-(1+o(1))I_n}.$$

When $\liminf_{n \rightarrow \infty} I_n / \log n > 1$, it exactly recovers all labels with probability tending to one. The initialization, selection, and refinement access only the retained eigenpairs. Each candidate fit may run any prescribed finite number of exact EM updates, with exact unconstrained or floored-simplex weight updates; no convergence to a global mixture maximum is assumed. A polynomial iteration bound gives a polynomial-time algorithm.

The second theorem retains the original growing fit as a candidate and proves the likelihood-residual safeguard. It does not claim that the growing branch without that safeguard has been proved for general K . The restart count $\lceil \log n \rceil$ needs no knowledge of P , its separation constants, or the desired confidence power.

3.6 A sufficient independent-row projection condition

The next theorem proves a usable bridge between subspace estimation and rare-event LLR accuracy. It is a result about a row-independent projection, not an unconditional guarantee for the full-graph fitted algorithm. Leave-one-out projections provide a natural setting for this analysis.

Theorem 3.10 (Row-independent spectral projection). *Fix i and restrict vectors and projectors to coordinates $j \neq i$. Let Π_i be a random orthogonal projector independent of $(A_{ij})_{j \neq i}$, and let $\Pi_{*,i}$ project onto the membership space of the other nodes. Suppose that on an event H_i , measurable with respect to the data generating Π_i ,*

$$\|(\text{Id} - \Pi_i)\Pi_{*,i}\|_{\text{op}} \leq \eta_n \rightarrow 0, \quad \max_{j \neq i} (\Pi_i)_{jj} \leq \frac{L}{n}, \quad (3.20)$$

for a constant L , and that $\Pr(H_i^c) \leq \gamma_n$. Define

$$L_{i,\Pi}^{ab} = A_{i,-i}\Pi_i w_{-i}^{ab} + c_i^{ab}. \quad (3.21)$$

There are deterministic $\tau_n = o(d_n)$ and $h_n \rightarrow \infty$ such that

$$\Pr \left\{ \max_{a \neq b} |L_{i,\Pi}^{ab} - L_i^{ab}| > \tau_n \right\} \leq \gamma_n + 2K^2 e^{-h_n d_n}. \quad (3.22)$$

Consequently, if $\liminf_{n \rightarrow \infty} (-\log \gamma_n) / I_n \geq 1$ uniformly in i , classifiers based on these oracle-parameter projected scores satisfy (3.14).

The proof uses a quadratic population bias and a Bennett tail bound. It requires bounded projector leverage, rather than a vanishing maximum coordinate error. Its independence assumption is explicit: forming $A_{i,-i}\Pi_i$ directly is not an additional step of our post-compression algorithm. To apply the theorem to that algorithm, one must further control the difference between its empirical full-graph projection and an independent-row construction, and then the remaining terms in (3.15). Lemma 3.5 and Theorem 3.6 provide a different full-graph route in logarithmic density. The independent-row result remains useful for representations outside that setting.

4 Extension below logarithmic density

4.1 Recovery target

Consider $d_n \rightarrow \infty$ with $d_n = o(\log n)$, keeping K , positive limiting community proportions, and the normalized probability matrix $(n/d_n)P$ fixed. The pairwise population profiles remain distinct, and the leading oracle information is proportional to d_n . The appropriate target is a vanishing fraction of errors with bound

$$\mathbb{E} \text{Mis}(\hat{z}, z) \leq e^{-(1+o(1))I_n}, \quad I_n \asymp d_n. \quad (4.1)$$

For consistency the right-hand side tends to zero. It generally does not yield exact recovery when $d_n = o(\log n)$. The distinction concerns the recovery objective, rather than a change in the meaning of the likelihood-ratio benchmark.

The score-transfer theorem in Section 3 applies at this density whenever its score-error conditions hold. The sparse Bernoulli-to-Poisson approximation remains valid because its exponent-scale correction is $O(d_n^2/n) = o(d_n)$. These facts provide a precise route for extension. They do not by themselves control the empirical eigenspace.

4.2 Representation and calibration

Low-degree concentration is more delicate: a few atypically high-degree vertices can produce localized empirical directions. Regularization can stabilize a spectral initializer, but a regularized representation must also retain the directions used for likelihood classification. Existing regularized spectral consistency results [11] concern recovery or concentration under their stated conditions; they do not automatically imply exponentially accurate spectral likelihood decoding.

For example, suppose a positive diagonal normalization D is used and the retained eigenpairs satisfy

$$M = D^{-1/2}AD^{-1/2}, \quad MV = V\Gamma, \quad \Pi_M = VV^\top. \quad (4.2)$$

The reconstruction in the original scale is

$$\tilde{A}_D = D^{1/2}V\Gamma V^\top D^{1/2} = AD^{-1/2}\Pi_M D^{1/2}. \quad (4.3)$$

Consequently $\tilde{A}_D w = Aw$ whenever $D^{1/2}w \in \text{col}(V)$. This exact identity exhibits the needed transformed contrast. The normalization factors must be retained with the eigenpairs to perform this reconstruction. A normalized embedding without its scale factors is not covered by this identity.

Trimming is a different operation. If $M = A - T$ is obtained by modifying entries, its retained reconstruction is $M\Pi_M$, and

$$A_i w - (M\Pi_M)_i w = T_i w + M_i(\text{Id} - \Pi_M)w. \quad (4.4)$$

Both the removed-edge contrast and projection error must be controlled at the recovery scale. This prevents interpreting every regularized spectral embedding as automatically equivalent to the original-graph LRT.

4.3 A conditional extension

Corollary 4.1 (Lower-density transfer). *Under the fixed normalized probability model with $d_n \rightarrow \infty$, $d_n/n \rightarrow 0$, and positive limiting proportions, let a spectral decoder, possibly calibrated as in (4.3), satisfy the score approximation and exceptional-event bounds of Theorem 3.4. Then its expected misclassification rate satisfies (4.1). If $d_n = o(\log n)$, this statement establishes almost exact recovery, without an exact-recovery conclusion.*

Proof. The Chernoff expansion and score-transfer proof use $d_n \rightarrow \infty$ and $d_n^2/n = o(d_n)$, the latter following from $d_n/n \rightarrow 0$. Substitute the density into Theorem 3.4. Distinct fixed normalized block rows and positive limiting proportions give $I_n \asymp d_n \rightarrow \infty$, so the expected fraction of errors tends to zero. $\square$

Verifying this corollary for a specified regularizer and fitting procedure remains a separate statistical task. The present 240-graph evaluation uses the unregularized logarithmic-density representation. Bounded expected degree, where detection has a different objective and threshold, is outside the scope of these results.

5 Experiments

The archived 240-graph comparisons evaluate the original growing branch of Section 2.3 and the separately specified controls. They do not evaluate the complete likelihood-restart Algorithm 1 on that full collection. The new safeguard is a proved algorithmic addition; its separate implementation verification is documented in the repository.

We evaluate whether likelihood-based decoding improves on a Euclidean decoder of the same retained eigenpairs, and whether global likelihood gains prevent redundant initialization. The experiments use a spectral profile working likelihood. They do not evaluate an exact leave-one-out graph likelihood or estimate an asymptotic error exponent from finite samples.

5.1 Models and protocol

For each model, let $C^{(0)}$ be a fixed symmetric connectivity matrix and let π be the vector of community proportions. We generate an undirected SBM without self-loops, with edge probabilities

$$P_{ab} = \frac{\log n}{n} C_{ab}, \quad C = \frac{\eta}{D_{\min}(C^{(0)}, \pi)} C^{(0)}, \quad (5.1)$$

where $\eta \in \{0.8, 1.4\}$ and

$$D_{\min}(C, \pi) = \min_{a \neq b} \max_{0 \leq t \leq 1} \sum_{c=1}^K \pi_c \left[(1-t) C_{ac} + t C_{bc} - C_{ac}^{1-t} C_{bc}^t \right]. \quad (5.2)$$

Thus the limiting minimum Chernoff–Hellinger coefficient equals η . The normalization uses the limiting proportions π ; actual group sizes are $\lfloor n\pi_a \rfloor$, with the remaining vertices assigned to the

last group. Vertex labels are then randomly permuted. Each model is tested at $n \in \{1000, 2000\}$. The six models are

$$\begin{aligned}
C_{\text{sym2}}^{(0)} &= \begin{pmatrix} 4 & 1 \\ 1 & 4 \end{pmatrix}, & \pi_{\text{sym2}} &= (1/2, 1/2), \\
C_{\text{equal3}}^{(0)} &= \begin{pmatrix} 10 & 1 & 2 \\ 1 & 11 & 1 \\ 2 & 1 & 10 \end{pmatrix}, & \pi_{\text{equal3}} &= (1/3, 1/3, 1/3), \\
C_{\text{hetero3}}^{(0)} &= \begin{pmatrix} 6 & 0.5 & 0.5 \\ 0.5 & 12 & 0.5 \\ 0.5 & 0.5 & 24 \end{pmatrix}, & \pi_{\text{hetero3}} &= (1/3, 1/3, 1/3), \\
C_{\text{unequal3}}^{(0)} &= \begin{pmatrix} 10 & 1 & 1 \\ 1 & 10 & 1 \\ 1 & 1 & 10 \end{pmatrix}, & \pi_{\text{unequal3}} &= (0.5, 0.3, 0.2), \\
C_{\text{hier4}}^{(0)} &= \begin{pmatrix} 12 & 2 & 0.5 & 0.5 \\ 2 & 12 & 0.5 & 0.5 \\ 0.5 & 0.5 & 12 & 2 \\ 0.5 & 0.5 & 2 & 12 \end{pmatrix}, & \pi_{\text{hier4}} &= (1/4, 1/4, 1/4, 1/4), \\
C_{\text{dis3}}^{(0)} &= \begin{pmatrix} 0.5 & 8 & 8 \\ 8 & 0.5 & 8 \\ 8 & 8 & 0.5 \end{pmatrix}, & \pi_{\text{dis3}} &= (1/3, 1/3, 1/3).
\end{aligned}$$

These include unequal group sizes, heterogeneous expected degrees, hierarchical connections, and negative signal eigenvalues.

The design batch contains five graph seeds, 20000, $\dots$, 20004, for every model, size, and signal coefficient, giving $6 \times 2 \times 2 \times 5 = 120$ graphs. A second batch uses the same models and hyperparameters with fresh seeds 30000, $\dots$, 30004. The algorithms and second-batch seed range were fixed before fitting the second batch. The second batch therefore validates initialization choices made using the first batch, rather than introducing new models. All methods are evaluated on identical eigenpairs within a graph.

We retain the K eigenpairs of the raw adjacency matrix with largest absolute eigenvalues. The implementation requests $K + 1$ eigenpairs using the symmetric ARPACK solver with tolerance 10^{-8} and discards the last one after sorting by absolute value. Neither spectral regularization nor degree rescaling is used. After this preprocessing, the fitting routines receive only (U, Λ) ; adjacency entries, degrees, true labels, and true block probabilities are unavailable to fitting and model selection.

5.2 Implementations and comparisons

The archived implementation forms positive spectral profiles

$$q_{ij} = \max \left\{ (U \Lambda U^\top)_{ij}, 10^{-4} \frac{\log n}{n} \right\}, \quad \ell_i(\hat{p}) = \sum_{j=1}^n q_{ij} \log \hat{p}_j - \sum_{j=1}^n \hat{p}_j. \quad (5.3)$$

The reconstructed diagonal is retained, even though the generated graphs have zero diagonal. Thus this is a declared positive-profile working score, not the original Bernoulli graph likelihood.

Component means are unrestricted n -dimensional profile means. Every likelihood arm uses the same soft EM updates and observed working objective

$$\mathcal{L}(\hat{\pi}, \hat{p}) = \sum_{i=1}^n \log \left[\sum_{a=1}^K \hat{\pi}_a \exp\{\ell_i(\hat{p}_a)\} \right]. \quad (5.4)$$

EM initializes equal component weights except when refitting a replacement proposal, where the current weights are retained. The mean update is the responsibility-weighted average of q_i . The weight update exactly maximizes the EM auxiliary objective on $\{\hat{\pi}_a \geq 10^{-8}, \sum_a \hat{\pi}_a = 1\}$. No additional floor on means is needed because the observations are positive. Final labels maximize $\log \hat{\pi}_a + \ell_i(\hat{p}_a)$.

The original deterministic growing branch of Section 2.3 is recorded as `growing_global_gain_EM`. It starts from the global profile mean, adds $K - 1$ node candidates by their all-node marginal gains, and runs joint EM after each addition. It requires no top-subset size and removes no nodes.

The frozen-profile control, recorded as `global_gain_no_update`, starts with the historical top-subset rule and then chooses all remaining node seeds by all-node gains, keeping the profiles frozen until one final EM fit. Its exact historical subset sizes are recorded in Appendix C. It is a different algorithm despite its identical recovery errors in these experiments.

We compare it with the following alternatives. *Top-subset peeling* uses the historical first-seed rule and repeatedly removes the selected candidate and its best-scoring subset, using the archived sizes specified in the appendix. After K selections, all n observations, including unselected nodes, enter EM. *Peeling plus replacement* fits this initialization, deletes each component temporarily, proposes the globally most useful node against the remaining $K - 1$ means, and refits all nodes. All K proposals use the same original fitted endpoint; the best proposal is accepted only if its final $\mathcal{L}$ exceeds the current best by more than 10^{-5} . This is one replacement pass, not repeated optimization.

The API option `method='global_gain'` invokes the primary growing algorithm; it does not invoke the frozen-profile control. *Residual++* samples the first node uniformly and later nodes proportionally to

$$d_i = \min_{a \in S} \sum_j [q_{ij} \log(q_{ij}/\hat{p}_{aj}) - q_{ij} + \hat{p}_{aj}]. \quad (5.5)$$

This deviance is not squared again. Its greedy variant samples $2 + \lfloor \log K \rfloor$ candidate trials per addition and chooses the trial that minimizes total residual deviance. We report both one start and ten starts; the ten-start fit is selected solely by final $\mathcal{L}$. An additional control selects one set of Euclidean k -means++ seeds in $X = \sqrt{n}U\Lambda$ and then runs the same Poisson-profile EM. The main Euclidean baseline runs k -means in X with ten starts, at most 300 Lloyd iterations per start, and tolerance 10^{-5} .

All profile EM fits use at most 300 updates and tolerance 10^{-6} . The stopping rule requires unchanged hard labels, maximum relative mean or absolute weight change at most 10^{-3} , and objective increment at most $10^{-6}(1 + |\mathcal{L} - G|)$ in absolute value, where $G = \sum_{ij} \log \Gamma(q_{ij} + 1)$ is a cached observation-only constant. This constant sets the stopping scale and does not affect fitting or selection. Randomized seeding uses fixed NumPy seed sequences indexed by graph seed, method, and restart. Reported error rates minimize label disagreement over all permutations of the K labels. Averages weight graphs equally; “exact” counts graphs with no mislabeled node. True labels are accessed only after all fitting and likelihood-based selections finish.

Table 1: Average mislabeled fraction in percent; exact recovery counts are out of 120 graphs per batch. All profile arms use the same EM score.

Method	Design batch		Independent batch	
	Error (%)	Exact	Error (%)	Exact
Top-subset peeling + EM	0.3917	86	0.3725	77
Peeling + replacement	0.0967	86	0.0925	77
Frozen global gain + EM	0.0967	86	0.0925	77
Growing global gain	0.0967	86	0.0925	77
Growing sampled gain + EM	0.3900	86	0.0925	77
Residual++, one start + EM	4.0129	80	3.1675	72
Greedy residual++, one start + EM	1.1192	85	2.3438	74
Greedy residual++, ten starts + EM	0.0967	86	0.0925	77
Euclidean++, one start + EM	3.9033	78	6.4204	71
X k -means, ten starts	0.3454	73	0.2933	66

Table 2: Both batches combined, 40 graphs per model. The main columns use the growing global-gain branch in the original growing branch. The frozen-profile control has identical recorded error counts.

Model	Growing global gain + EM		X k -means	
	Error (%)	Exact	Error (%)	Exact
Symmetric two-community	0.0150	33	0.0125	34
Equal-size three-community	0.0175	32	0.0188	31
Heterogeneous three-community	0.1625	18	1.4175	1
Unequal-size three-community	0.0338	33	0.1213	25
Hierarchical four-community	0.0275	28	0.0288	29
Disassortative three-community	0.3113	19	0.3175	19

5.3 Recovery results

Table 1 gives the paired results in the design and independent batches. Global-gain initialization, peeling plus replacement, growing global gain, and ten-start greedy residual++ have the same per-graph error counts in both batches. Across all 240 graphs, the main deterministic method has average error 0.0946%, compared with 0.3194% for the ten-start Euclidean baseline, and exactly recovers 163 graphs compared with 139. These are finite-sample recovery comparisons with unequal optimization budgets, rather than equal-runtime comparisons.

Table 2 separates model families. The largest improvement over Euclidean decoding occurs with heterogeneous diagonal connectivities: the average error falls from 1.4175% to 0.1625%. The likelihood criterion also improves the unequal-size model. It is not uniformly better: the symmetric two-community baseline has one fewer mislabeled node in aggregate, and the hierarchical model has one additional exactly recovered graph under k -means. The examples therefore support likelihood adaptation when block profiles differ substantially, without claiming dominance on every graph.

The stronger signal coefficient $\eta = 1.4$ gives 117 exactly recovered graphs out of 120 for global gain, versus 99 for k -means. At $\eta = 0.8$, the corresponding counts are 46 and 40. This behavior is consistent with the importance of the logarithmic-density recovery threshold, but the experiment does not establish that threshold or the leading error exponent.

5.4 A failure of subset scores and its repair

One design graph with $n = 2000$, $K = 3$, and $\eta = 0.8$ illustrates the initialization mechanism. Its normalized connectivity matrix is approximately

$$C = \begin{pmatrix} 7.8541 & 0.7854 & 1.5708 \\ 0.7854 & 8.6395 & 0.7854 \\ 1.5708 & 0.7854 & 7.8541 \end{pmatrix}, \quad (5.6)$$

with group sizes (666, 666, 668). Communities 1 and 3 are the closest pair: their empirical X -centroid distance is 40.22, compared with 47.36 and 47.82 for the other pairs. They are nevertheless well separated in the retained spectrum. Using their true profile means as a diagnostic classifier assigns every node correctly on its first assignment.

Top-subset peeling selects nodes 858, 610, 435 (zero-based IDs), whose true communities are 2, 3, 2. The profile of node 610 lies near the mixture of communities 1 and 3. Its Poisson divergence to their merged mean is 0.3826, compared with 5.2433 to its own community's mean. Its selected subset has 150 nodes from community 1 and 196 from community 3. Thus a candidate can win a subset likelihood comparison by representing a mixture of communities. The resulting fit merges communities 1 and 3 and splits community 2, making 709 errors (35.45%). Tightening the tolerance to 10^{-10} and permitting 3000 updates does not solve this structural problem: the fit stops after 517 updates with 800 errors.

The frozen global-gain control keeps the first two seeds but changes the third to node 220, which contributes to the uncovered region; after EM, only one node is mislabeled. A one-component replacement at the original fitted endpoint selects the same node and likewise reduces errors to one, while increasing the working log likelihood by approximately 7148.48. No true labels enter either choice. The independent batch contains a second analogous peeling failure with 675 errors, reduced to three by replacement. Relative to peeling, each effective deterministic variant improves exactly one graph in each batch, leaves the other 119 error counts unchanged, and worsens none. The number of exact recoveries does not change because the repaired graphs retain one and three errors, respectively.

5.5 Numerical verification and computational scope

All returned profile fits satisfy the declared stopping rule in the 240-graph comparison. The 120 original peeling errors are reproduced exactly. The archive contains 6240 EM traces, including replacement proposals, growing stages, and restart fits, with 27867 recorded trace points. The smallest observed objective increment is -1.17×10^{-10} , within floating-point rounding; no increment below -10^{-7} occurs. All 240 selected ten-start solutions maximize the same final working likelihood within their restart set. Replacement accepts a likelihood improvement on 22 design graphs and 25 independent graphs, and never lowers the final objective; most of these small improvements do not change the classification error. Independent scalar checks reproduce deviance and global-gain calculations with maximum absolute errors 7.11×10^{-15} and 1.78×10^{-15} , respectively.

The present implementation explicitly reconstructs the $n \times n$ profiles and caches all singleton-candidate scores. Reconstruction costs $O(n^2K)$, score precomputation costs $O(n^3)$, and storage costs $O(n^2)$. An EM update costs $O(n^2K)$. A replacement pass tries K additional fits, whereas ten-start residual seeding uses ten fits. The runs use one numerical library thread per worker and four worker processes. Python, NumPy, SciPy, and scikit-learn versions are 3.11.7, 2.1.3, 1.16.2, and 1.7.1. Stored timers cover all candidate arms and restarts together and exclude graph generation

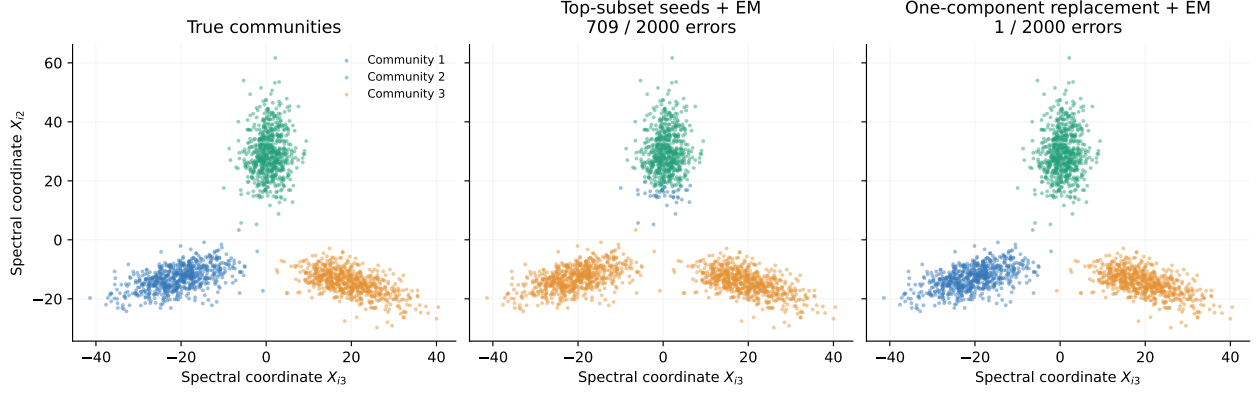


Figure 1: The saved design failure, displayed in two coordinates of $X = \sqrt{n}U\Lambda$. Left: true communities, used only for evaluation. Middle: top-subset peeling followed by EM merges the two lower communities and splits the upper community. Right: an accepted one-component replacement followed by EM separates the groups and leaves one mislabeled node. Predicted labels are aligned to truth for display only.

and eigenpair computation, so they do not support an arm-specific speed claim. The data sizes $n \leq 2000$ establish feasibility for this implementation; candidate-pool approximations and blockwise computations require separate large-graph validation.

The evidence supports improved decoding of a fixed spectrum and a repair of redundant initialization under the declared working likelihood. It does not by itself prove that the practical EM endpoint has the oracle error exponent, that clipping preserves exact graph likelihood ratios, or that the growing initializer meets the high-probability weak-start condition in Theorem 3.6. Oracle-LRT comparisons, a denser range of n for error-exponent estimation, and lower-density regularized experiments remain outside the archived numerical study.

6 Conclusion

Spectral compression can preserve the likelihood information needed for nearly optimal community recovery. In a full-rank SBM, the membership subspace contains the block-constant likelihood contrasts. First-order analysis of the empirical adjacency eigenpairs makes this relation quantitative at logarithmic density, and an admissible fixed positive floor preserves the leading Chernoff exponent. The subsequent likelihood and EM analysis controls the dependent whole-graph profiles.

For $K = 2$, the original growing likelihood procedure has an end-to-end guarantee. For general fixed K , Algorithm 1 supplements the growing fit with $\lceil \log n \rceil$ independent likelihood-residual++ initializations, compares the fitted candidates using the same observed working mixture likelihood, and performs a final EM update before decoding. The resulting recovery bound is

$$\mathbb{E} \text{Mis}(\hat{z}, z) \leq e^{-(1+o(1))I_n}.$$

Initialization and likelihood selection are part of the proof; weak recovery is not an assumption imposed on an unspecified initializer. The guarantee matches the leading oracle exponent and yields exact recovery above the corresponding logarithmic information threshold.

The paired 240-graph experiments evaluate the original growing and parameter-replacement procedures. They support the usefulness of all-node likelihood gains and likelihood-accepted repairs

in resolving severe initialization failures. A full evaluation of the safeguarded combination remains to be carried out. Removing the safeguard while retaining the general fixed- K theorem, establishing calibrated spectral concentration for a lower-density regularizer, and reducing the cost of full profile reconstruction are separate extensions of the completed logarithmic-density theory.

A Proofs of spectral preservation and statistical results

A.1 Exact preservation and the population contrast space

Proof of Proposition 3.1. Since $AU = U\Lambda$, multiplication by $U^\top$ gives $A\Pi = AUU^\top = U\Lambda U^\top = \tilde{A}$. Therefore $\tilde{A}_i w - A_i w = A_i(\Pi - \text{Id})w$. If $w \in \text{col}(U)$, then $\Pi w = w$ and the difference vanishes.

Let $N = Z^\top Z = \text{diag}(n_1, \dots, n_K)$, which is nonsingular since all communities are nonempty. The columns of $ZN^{-1/2}$ are orthonormal, and

$$ZPZ^\top = (ZN^{-1/2})(N^{1/2}PN^{1/2})(ZN^{-1/2})^\top.$$

Nonsingularity of P makes the central matrix nonsingular. Hence $ZPZ^\top$ has rank K and column space $\text{col}(Z)$. For each contrast, define $\xi_c^{ab} = \log\{P_{ac}(1 - P_{bc})/[P_{bc}(1 - P_{ac})]\}$. Then $w^{ab} = Z\xi^{ab}$, which proves membership in that space. Finally, subtracting $\text{diag}(P_{z_i z_i})$ from $ZPZ^\top$ gives the actual no-self-loop mean, so no rank assertion about that corrected matrix is needed. $\square$

Full rank cannot be replaced by distinct block rows without a further contrast-span condition. For example, let $P = (d_n/n)ss^\top$ for a positive, nonconstant vector s . Its signal space is spanned by Zs , whereas the Poisson contrast between communities a, b is $\log(s_a/s_b)\mathbf{1}$. When $s_a \neq s_b$, this constant vector does not belong to $\text{span}(Zs)$. A rank-one signal projection does not preserve that contrast by the preceding identity. This example only concerns projection onto the population rank-one space; it is not an impossibility claim for every statistic computed from one empirical eigenvector.

A.2 The sparse likelihood correction

Proof of Lemma 3.2. Write $p_{aj} = P_{a,z_j}$ and $p_{bj} = P_{b,z_j}$ for $j \neq i$. The exact binary Bernoulli log-likelihood ratio is

$$L_i^{ab} = \sum_{j \neq i} A_{ij} \log \frac{p_{aj}}{p_{bj}} + \sum_{j \neq i} A_{ij} \log \frac{1 - p_{bj}}{1 - p_{aj}} + \sum_{j \neq i} \log \frac{1 - p_{aj}}{1 - p_{bj}}.$$

For $0 \leq x \leq 1/2$, the power series for $\log(1-x)$ implies $|\log(1-x)+x| \leq Cx^2$. Since $p_{aj}, p_{bj} \leq b_+ d_n/n$, it also implies $|\log((1 - p_{bj})/(1 - p_{aj}))| \leq C d_n/n$ and

$$\left| \log \frac{1 - p_{aj}}{1 - p_{bj}} + (p_{aj} - p_{bj}) \right| \leq C d_n^2/n^2.$$

The first term in the exact LLR is the Poisson linear term, and the linear approximation to its final term is the Poisson offset. Summing the two displayed remainders proves (3.6).

Under any true community, $\deg(i)$ is a sum of independent Bernoulli variables with mean at most $b_+ d_n$. For $D > b_+$, exponential Markov inequality with $t = \log(D/b_+)$ and the bound $\log \mathbb{E} e^{t \deg(i)} \leq b_+ d_n (e^t - 1)$ give

$$\Pr\{\deg(i) > D d_n\} \leq \exp\{-d_n[D \log(D/b_+) - D + b_+]\}.$$

The bracket tends to infinity with D , so D_H can be chosen for any H . On the complementary event the correction is at most $C(D_H + 1)d_n^2/n = o(d_n)$. If $d_n = \log n$, this becomes $O(\log^2 n/n)$. $\square$

A.3 Chernoff information and the oracle exponent

Proof of Lemma 3.3. Let f_a, f_b denote the product Bernoulli mass functions of the incident-edge row in the two hypotheses. Define

$$\begin{aligned} D_\alpha(p_{a,-i} \| p_{b,-i}) &= -\log \sum_x f_a(x)^{1-\alpha} f_b(x)^\alpha \\ &= -\sum_{j \neq i} \log \{p_{aj}^{1-\alpha} p_{bj}^\alpha + (1-p_{aj})^{1-\alpha} (1-p_{bj})^\alpha\}, \\ D_{i,ab}^* &= \max_{0 \leq \alpha \leq 1} D_\alpha(p_{a,-i} \| p_{b,-i}). \end{aligned}$$

Uniformly for $\alpha \in [0, 1]$ and $u, v \in [b_- d_n/n, b_+ d_n/n]$,

$$u^{1-\alpha} v^\alpha + (1-u)^{1-\alpha} (1-v)^\alpha = 1 - \{(1-\alpha)u + \alpha v - u^{1-\alpha} v^\alpha\} + O(d_n^2/n^2).$$

For example, the nonedge product expansion follows by writing it as $\exp\{(1-\alpha)\log(1-u) + \alpha\log(1-v)\}$; the first and second derivatives of the nonedge factors are uniformly bounded for $u, v \leq 1/2$. Expanding $-\log(1-x)$ and summing gives

$$D_{i,ab}^* = n \max_{0 \leq \alpha \leq 1} \sum_c \pi_{c,n} \{(1-\alpha)P_{ac} + \alpha P_{bc} - P_{ac}^{1-\alpha} P_{bc}^\alpha\} + O(d_n^2/n + d_n/n), \quad (\text{A.1})$$

where $\pi_{c,n} = n_c/n$. The $O(d_n/n)$ term accounts for the excluded node. Since $(n/d_n)P$ is fixed and $\pi_{c,n} \rightarrow \pi_c > 0$, the displayed expression divided by d_n has a finite limit. Distinct normalized block rows make this limit positive: taking $\alpha = 1/2$ gives one half the positive weighted sum of the squared differences of their square roots. Thus $D_{i,ab}^* \asymp d_n$ uniformly over the finitely many pairs and excluded nodes, and $I_n \asymp d_n$.

The equal-prior Bayes risk is $R_{i,ab}^{\text{or}} = \frac{1}{2} \sum_x \min\{f_a(x), f_b(x)\}$. Since $\min\{u, v\} \leq u^{1-\alpha} v^\alpha$,

$$R_{i,ab}^{\text{or}} \leq \frac{1}{2} e^{-D_{i,ab}^*}. \quad (\text{A.2})$$

For completeness, a logarithmic lower bound follows by exponential tilting, without invoking any refined rate formula. Let α_* maximize $D_\alpha(p_{a,-i} \| p_{b,-i})$ and set

$$f_*(x) = e^{D_{i,ab}^*} f_a(x)^{1-\alpha_*} f_b(x)^{\alpha_*}, \quad L(x) = \log \frac{f_a(x)}{f_b(x)}.$$

All masses are positive, the two distributions differ, and the endpoint derivatives have opposite signs. Thus $\alpha_* \in (0, 1)$, and differentiating the normalization gives $\mathbb{E}_* L = 0$. The tilted incident edges remain independent, with probabilities

$$p_{*,j} = \frac{p_{aj}^{1-\alpha_*} p_{bj}^{\alpha_*}}{p_{aj}^{1-\alpha_*} p_{bj}^{\alpha_*} + (1-p_{aj})^{1-\alpha_*} (1-p_{bj})^{\alpha_*}} \leq C d_n/n.$$

The coefficient of each edge in L is bounded by a constant depending only on b_-, b_+ ; hence $\text{Var}_*(L) \leq C d_n$. Taking $h_n = d_n^{2/3}$, Chebyshev's inequality gives $\Pr_*(|L| \leq h_n) = 1 - o(1)$. The change of measure yields

$$\begin{aligned} R_{i,ab}^{\text{or}} &= \frac{1}{2} e^{-D_{i,ab}^*} \mathbb{E}_* \min\{e^{\alpha_* L}, e^{-(1-\alpha_*)L}\} \\ &\geq \frac{1}{2} e^{-D_{i,ab}^* - h_n} \Pr_*(|L| \leq h_n). \end{aligned}$$

Combining this with (A.2) proves $\log R_{i,ab}^{\text{or}} = -D_{i,ab}^* + o(d_n)$, and hence (3.10). For priors $\nu, 1 - \nu$ bounded away from zero, $\min\{\nu f_a, (1 - \nu)f_b\}$ is bounded above and below by constant multiples of $\min\{f_a, f_b\}$, so its leading exponent is unchanged. $\square$

A.4 Transfer of the leading exponent

Proof of Theorem 3.4. Suppose the true label of node i is a . If G_i occurs and the classifier prefers $b \neq a$, then $\hat{\ell}_i(a) - \hat{\ell}_i(b) \leq 0$, and consequently $L_i^{ab} \leq \tau_n$. For every $\alpha \in [0, 1]$,

$$\Pr\{L_i^{ab} \leq \tau_n\} \leq e^{\alpha\tau_n} \mathbb{E}_a e^{-\alpha L_i^{ab}} = \exp\{\alpha\tau_n - D_\alpha(p_{a,-i} \| p_{b,-i})\}.$$

Selecting a Chernoff maximizer and using $\alpha \leq 1$ gives

$$\Pr\{\hat{z}_i \neq z_i\} \leq \delta_n + \sum_{b \neq a} \exp\{\tau_n - D_{i,ab}^*\} \leq \delta_n + (K - 1) \exp\{-I_n + o(d_n)\}.$$

Since $I_n \asymp d_n$, we have $o(d_n) = o(I_n)$, and (3.13) makes the first term no larger in leading exponential order. Averaging over nodes and using that the permutation-invariant error is no larger than the error under any one common alignment proves (3.14).

The expected number of errors under that alignment is at most $n \exp\{-(1 + o(1))I_n\}$. If $\liminf_{n \rightarrow \infty} I_n / \log n > 1$, Markov's inequality gives vanishing probability of even one error, which proves the exact recovery assertion. No conclusion when $I_n / \log n \rightarrow 1$ follows from this argument. $\square$

The matching binary lower bound mentioned after the theorem has a direct decision-theoretic justification. Conditional on all other labels, the rest of the graph is independent of the incident-edge row and has the same distribution under a and b . Revealing the other labels and true parameters cannot increase the Bayes risk. A function of the graph's spectral summary is also a function of the full graph; its risk in that binary experiment is therefore at least $R_{i,ab}^{\text{or}}$. This argument does not, by itself, assert a global minimax lower bound over an unspecified parameter space.

A.5 Full-graph reconstruction and local empirical EM

Proof of Lemma 3.5. The true no-self-loop mean $\mathbb{E}[A \mid z]$ leaves $\text{col}(Z)$ invariant. In the orthonormal membership basis $ZN^{-1/2}$ its restriction is $N^{1/2}PN^{1/2} - \text{diag}(P_{11}, \dots, P_{KK})$. Nonsingularity of the fixed matrix $(n/d_n)P$ and positive limiting proportions imply that its K eigenvalues have absolute values between $\kappa_0 d_n$ and Cd_n for some $\kappa_0 > 0$ and all large n . On the orthogonal complement, the eigenvalues are $-P_{aa} = O(d_n/n)$. Let U_* and Λ_* contain these K signal eigenpairs; then $U_* U_*^\top = \Pi_*$ and $\|U_*\|_{2 \rightarrow \infty} = O(n^{-1/2})$.

Here $\|M\|_{2 \rightarrow \infty} = \max_i \|M_i\|_2$. We spell out how the cited entrywise theorem applies. Fix any desired polynomial confidence power. Lei–Rinaldo's spectral norm theorem [7, Theorem 5.2], with its confidence power chosen arbitrarily large, gives $\|A - \mathbb{E}[A \mid z]\|_{\text{op}} \leq C_H \sqrt{d_n}$ outside $O(n^{-H})$. Its upper degree parameter can be taken as $\max\{1, b_+\}d_n$. The signal separation is of order d_n , and the eigenvalue condition number is bounded. In the notation of Abbe et al. [2, Theorem 2.1], take $\gamma = C_H / \sqrt{d_n}$. Their incoherence assumption holds because $\|\mathbb{E}[A \mid z]\|_{2 \rightarrow \infty} = O(d_n / \sqrt{n})$; their row/column independence assumption follows from the independent upper-triangular edges.

For row concentration, their Bernoulli Lemma 7 applies with upper probability $b_+ d_n / n$. Choosing its tail parameter large enough gives row failure $O(n^{-H-2})$. For matrices with a fixed number of

columns, applying it to the columns and using monotonicity of φ_H and of $\varphi_H(x)/x$ to replace the column norms by the Frobenius and maximum-row norms gives the required function $\varphi_H(x) = C_H/(1 \vee \log(1/x))$ for $x > 0$, with $\varphi_H(0) = 0$. The constants absorb the fixed number of columns and the ratio of the signal gap to d_n . This function is continuous and nondecreasing, $\varphi_H(x)/x$ is nonincreasing, and both γ and $\varphi_H(\gamma)$ tend to zero. Thus the remaining smallness condition in that theorem holds for all sufficiently large n .

Apply the theorem separately to the positive and negative signal groups, each a contiguous group of eigenvalues, and concatenate their orthogonal alignments. Repeated eigenvalues within either group cause no difficulty. The theorem allows a non-low-rank mean, so the excluded diagonal has not been silently discarded. The norm event also ensures that the empirical largest-absolute-value eigenpairs are precisely these signal groups. We obtain an orthogonal R , $V = UR$, and deterministic $\nu_n \rightarrow 0$ such that

$$\|V\|_{2 \rightarrow \infty} \leq C_H/\sqrt{n}, \quad \|V - AU_*\Lambda_*^{-1}\|_{2 \rightarrow \infty} \leq \nu_n/\sqrt{n}.$$

The same polar alignments satisfy the Davis–Kahan bound $\|V - U_*\|_{\text{op}} \leq C_H/\sqrt{d_n}$. Put $M = R^\top \Lambda R = V^\top AV$. Since $\|\mathbb{E}[A \mid z]\|_{\text{op}} = O(d_n)$,

$$\|M - \Lambda_*\|_{\text{op}} \leq \|A - \mathbb{E}[A \mid z]\|_{\text{op}} + 2\|\mathbb{E}[A \mid z]\|_{\text{op}}\|V - U_*\|_{\text{op}} \leq C_H\sqrt{d_n}.$$

A degree union bound, with the degree-tail constant chosen for the same confidence power, gives $\max_i \deg(i) \leq C_H d_n$.

Set $E = V - AU_*\Lambda_*^{-1}$. The exact row decomposition is

$$(\tilde{A} - A\Pi_*)_i = E_i M V^\top + (AU_*)_i \{\Lambda_*^{-1} M V^\top - U_*^\top\}.$$

The bracket has operator norm $O(d_n^{-1/2})$, while $\|(AU_*)_i\|_2 \leq \deg(i)\|U_*\|_{2 \rightarrow \infty} \leq C_H d_n/\sqrt{n}$. Consequently

$$\|\tilde{A} - A\Pi_*\|_{2 \rightarrow \infty} \leq C_H(d_n \nu_n + \sqrt{d_n})/\sqrt{n} = o(d_n/\sqrt{n}).$$

Cauchy–Schwarz yields the required uniform row ℓ_1 bound. Entrywise clipping is 1-Lipschitz, so the bound also holds for $q - q^0$. Moreover, $|\tilde{A}_{ij}| \leq \|U_i\|_2 \|\Lambda\|_{\text{op}} \|U_j\|_2 \leq C_H d_n/n$, proving the profile upper bound.

For the average approximation, define the block counts $N_{ic} = \sum_{j: z_j=c} A_{ij}$. Because $c_0 d_n/n < P_{ac}$, clipping toward P_{ac} cannot increase the coordinate error:

$$\|q_i^0 - p_{z_i,*}\|_1 \leq \sum_c |N_{ic} - n_c P_{z_i,c}|.$$

Let F be the sum on the right over all i . Its expectation is at most $Cn\sqrt{d_n} + Cd_n$; the latter term accounts for the missing diagonal. Changing one independent edge changes F by at most two. The bounded-differences inequality therefore gives

$$\Pr\{F - \mathbb{E}F > nd_n^{3/4}\} \leq e^{-cd_n^{3/2}}.$$

In logarithmic density this is smaller than any fixed inverse polynomial. Combining this average bound with the uniform $q - q^0$ bound, and enlarging a deterministic vanishing a_n , proves all assertions. $\square$

Clipped oracle comparisons. For later use, fix a true class a and a competitor b . Write $v_c = \log(P_{ac}/P_{bc})$, $N'_{ic} = \max\{N_{ic}, n_c c_0 d_n/n\}$, and

$$L_{i,\text{clip}}^{ab} = \sum_c N'_{ic} v_c - \sum_c n_c (P_{ac} - P_{bc}).$$

This is a comparison score for the analysis, not a computable oracle supplied to the algorithm. For any $\alpha \in [0, 1]$, a coordinate with $v_c \geq 0$ has a clipped exponential moment no larger than its unclipped moment. If $v_c < 0$, put $\beta = -\alpha v_c \geq 0$ and $t_c = n_c c_0 d_n/n$. For all sufficiently large n , $t_c \leq \mathbb{E}_a N_{ic}$, including the excluded self-loop, because $c_0 < \min_{a,c} n P_{ac}/d_n$. Jensen's inequality implies

$$\mathbb{E}_a e^{\beta \max\{N_{ic}, t_c\}} \leq \mathbb{E}_a e^{\beta N_{ic}} + e^{\beta t_c} \leq 2 \mathbb{E}_a e^{\beta N_{ic}}.$$

The counts over different column blocks are independent for a fixed node. Thus clipping costs at most a factor 2^K in the Chernoff moment. A direct binomial moment expansion, including an $O(d_n/n)$ diagonal correction, gives

$$\log \mathbb{E}_a e^{-\alpha L_{i,\text{raw}}^{ab}} = -D_\alpha(p_{a,-i} \| p_{b,-i}) + O(d_n^2/n + d_n/n),$$

where the raw score replaces N'_{ic} by N_{ic} . For any deterministic $\tau_n = o(d_n)$, exponential Markov inequality at the maximizing α yields

$$\Pr\{L_{i,\text{clip}}^{ab} \leq \tau_n\} \leq 2^K \exp\{-D_{i,ab}^* + o(d_n)\}. \quad (\text{A.3})$$

The argument controls the error probability directly. It does not assume that clipping changes every rare-row score by $o(d_n)$.

Proof of Theorem 3.6. Choose the confidence power $H > 2 + \limsup_{n \rightarrow \infty} I_n / \log n$ in Lemma 3.5 and work on its event. Constants below may depend on the fixed model and floor but not on the iteration. Suppose a responsibility matrix R has aligned average error at most γ , with γ below a sufficiently small fixed constant. Put $N_a = \sum_i R_{ia}$ and $\bar{q}_a = n_a^{-1} \sum_{i:z_i=a} q_i$. Then $|N_a - n_a| \leq n\gamma$ and $N_a \geq \pi_{\min} n/2$, where $\pi_{\min}$ is a fixed lower bound on the community proportions for large n . The average-profile bound gives $\|\bar{q}_a - p_{a*}\|_1 \leq C a_n d_n$. The M-step identity gives, deterministically,

$$\|\hat{p}_a - \bar{q}_a\|_1 \leq \frac{\sum_i |R_{ia} - Z_{ia}| \|q_i\|_1}{N_a} + \left| \frac{n_a}{N_a} - 1 \right| \|\bar{q}_a\|_1 \leq C\gamma d_n.$$

Hence

$$\max_a \|\hat{p}_a - p_{a*}\|_1 \leq C(\gamma + a_n) d_n, \quad c_0 d_n/n \leq \hat{p}_{aj} \leq C_H d_n/n. \quad (\text{A.4})$$

The logarithm is Lipschitz on this common positive range, so its vector ℓ_1 error is at most Cn/d_n times the profile error. Since $q_{ij} \leq C_H d_n/n$, both the fitted linear contrasts and their offsets differ from the true-profile working contrasts by at most $C(\gamma + a_n) d_n$. The fitted log-weight ratios are bounded: in the unconstrained update N_a/n is bounded away from zero; for a fixed positive floor this follows directly from feasibility.

Combining the uniform row $q - q^0$ bound with these parameter bounds gives, for every node and pair,

$$|\hat{\ell}_i(a) - \hat{\ell}_i(b) - L_{i,\text{clip}}^{ab}| \leq C(\gamma + a_n) d_n + C. \quad (\text{A.5})$$

No independence of R , q , and the incident edges has been used. With $\gamma = \gamma_n \rightarrow 0$, a wrong classification implies $L_{i,\text{clip}}^{ab} \leq o(d_n)$ for some competitor, apart from the initialization or graph exceptional events. Equation (A.3) and a finite union bound prove the one-update rate.

We also verify that further EM updates remain in the required region. Let

$$\kappa = \liminf_n \min_{a \neq b} d_n^{-1} D_P(p_{a*} \| p_{b*}) > 0.$$

Choose a small fixed basin radius $\gamma_0 > 0$ so that the error bound from (A.4) for $\gamma \leq \gamma_0$ is less than $\kappa d_n/8$ in each score comparison for large n . Also choose a fixed small $t_0 > 0$ so that a row with $\|q_i - p_{z_i,*}\|_1 \leq t_0 d_n$ has an observation-contrast error less than $\kappa d_n/8$. At the population observation $q_i = p_{z_i,*}$, the working margin equals $D_P(p_{z_i,*} \| p_{b*}) \geq \kappa d_n/2$ for large n . The specified row deviation costs at most $\kappa d_n/8$, and the fitted-parameter and bounded weight corrections together cost at most $\kappa d_n/8$, leaving a correct-class margin of at least $\kappa d_n/4$. The average-profile bound shows that the remaining rows constitute at most a_n/t_0 of all nodes. Thus the next E-step has aligned average responsibility error at most

$$b_n = 2a_n/t_0 + 2(K-1)e^{-\kappa d_n/4} \rightarrow 0.$$

This bound holds uniformly for every R in the fixed basin. Eventually $b_n < \gamma_0$, so the basin is invariant. Starting with error $\gamma_n \rightarrow 0$, every later responsibility matrix has error at most $\max\{\gamma_n, b_n\} = o(1)$. Equation (A.5) therefore holds with one deterministic $o(d_n)$ bound for every later fitted score on the same event. This includes a data-dependent finite stopping time and needs no union bound over iterations.

Finally add the initialization failure δ_n and the graph failure $O(n^{-H})$ to the per-node Chernoff bound and average over nodes. Since $d_n = \log n$ and $H > 2 + \limsup_{n \rightarrow \infty} I_n / \log n$, the graph failure is negligible compared with e^{-I_n} , and the assumed initialization reliability preserves the leading exponent. If $\liminf_{n \rightarrow \infty} I_n / \log n > 1$, multiplying the expected error by n and applying Markov's inequality gives exact recovery with probability tending to one. $\square$

Proof of Corollary 3.7. The spectral norm and Davis–Kahan bounds above give an orthogonal alignment with $\|U - U_* R^\top\|_F^2 = O(d_n^{-1})$ outside $O(n^{-H})$. The K distinct population row centers have squared separation at least $2/n$, since the rows of $ZN^{-1/2}$ from different groups are orthogonal, with squared norms $1/n_a$ and $1/n_b$. A fixed-factor approximate K -means solution has total squared error at most a constant times $\|U - U_* R^\top\|_F^2$. The deterministic center-matching argument (also used in Lei–Rinaldo's clustering bound) then gives $\text{Mis}(\tilde{z}, z) = O(d_n^{-1})$: the total squared discrepancy of fitted centers from the population rows is $O(d_n^{-1})$; positive community sizes force one fitted center near each distinct population center; each incorrectly assigned row contributes a constant multiple of $1/n$ to that discrepancy. Hard responsibilities consequently have aligned average error $2 \text{Mis}(\tilde{z}, z) = O(d_n^{-1})$.

For a concrete randomized implementation, the K -means++ expected squared-error bound is $C_K = 8(\log K + 2)$ times the optimum [4]. Conditional on U , Markov's inequality bounds the chance of exceeding $2C_K$ times the optimum by $1/2$ for each independent seeding. Taking at least $(H+2)\log_2 n$ independent seedings and selecting the smallest squared-error potential makes that failure at most n^{-H-2} . Any subsequent Lloyd steps can only lower this potential. The zero-optimum case is recovered exactly by the seeding. For fixed K this costs polynomial time and uses only U . Choose $H > \limsup_{n \rightarrow \infty} I_n / \log n$, combine this event with the spectral event, and apply Theorem 3.6. $\square$

A.6 End-to-end likelihood initialization

Proof of Theorem 3.8. Fix $H > 2 + \limsup_{n \rightarrow \infty} I_n / \log n$ and work on the event of Lemma 3.5. All constants below may depend on H and the fixed model. Enlarge its deterministic $a_n \rightarrow 0$ if

necessary to bound both displayed profile errors. Put $\pi_{a,n} = n_a/n$,

$$m_b = \sum_{a=1}^2 \pi_{a,n} P_{ab}, \quad D_{\pi_n}(x||y) = \sum_b \pi_{b,n} \{x_b \log(x_b/y_b) - x_b + y_b\}.$$

The block values of q_c^0 are

$$v_{cb} = \max \left\{ \frac{\sum_{j:z_j=b} A_{cj}}{n_b}, \frac{c_0 d_n}{n} \right\}, \quad q_{cj}^0 = v_{c,z_j}.$$

This is only an analysis variable. In particular v_c may be a rare extreme candidate, and we do not assume it is near its own community profile. Let

$$h_a(v) = \frac{n}{d_n} \sum_b \pi_{b,n} \{P_{ab} \log(v_b/m_b) - (v_b - m_b)\}, \quad \Phi_n(v) = \sum_a \pi_{a,n} [h_a(v)]_+.$$

The global profile mean $\bar{q}$ differs from $(m_{z_j})_{j=1}^n$ in ℓ_1 norm by at most $a_n d_n$. The uniform bound on $q_c - q_c^0$ and the common positive box give

$$|[\ell_i(q_c) - \ell_i(\bar{q})] - d_n h_{z_i}(v_c)| \leq C\{\|q_i - p_{z_i,*}\|_1 + a_n d_n\}.$$

Here the parameter perturbation costs $C\|q_c - q_c^0\|_1 + C\|\bar{q} - (m_{z_j})_{j=1}^n\|_1$, because $q_{ij} \leq C d_n/n$ and log is $n/(c_0 d_n)$ -Lipschitz. The remaining observation perturbation multiplies a uniformly bounded log ratio. Since positive part is 1-Lipschitz, summing yields the simultaneous candidate bound

$$\sup_c \left| \frac{\Delta_2(c)}{n d_n} - \Phi_n(v_c) \right| \leq C a_n.$$

Every true class has a candidate with $\|q_c - p_{a*}\|_1 = o(d_n)$ by the average error bound and its positive proportion. The same is true of q_c^0 , so $(n/d_n) \sum_b \pi_{b,n} |v_{cb} - P_{ab}| = o(1)$. The fixed normalized matrix and positive limiting proportions imply that

$$g_0 = \lim_{n \rightarrow \infty} \max_a \frac{n \pi_{a,n}}{d_n} D_{\pi_n}((P_{ab})_{b=1}^2 \|m) > 0.$$

Positivity follows because the two distinct normalized block rows cannot both equal their mean. Continuity on the normalized common positive box and the candidate bound imply that the selected candidate satisfies $\Phi_n(v_c) \geq g_0/2$ for large n .

The decisive two-class identity is

$$\sum_a \pi_{a,n} h_a(v) = -\frac{n}{d_n} D_{\pi_n}(m||v) \leq 0.$$

Together with $\Phi_n(v_c) \geq g_0/2$, it forces exactly one of $h_1(v_c), h_2(v_c)$ to be positive. Moreover

$$h_{\text{pos}}(v_c) \geq g_0/2, \quad h_{\text{neg}}(v_c) \leq -g_0/2,$$

since each weighted positive or negative magnitude is at least $g_0/2$ and each proportion is at most one. The initial two mixture weights are equal, so they do not alter this score comparison.

Take $\tau_n = \sqrt{a_n} + d_n^{-1}$. At most $(a_n/\tau_n)n = o(n)$ observations have $\|q_i - p_{z_i,*}\|_1 > \tau_n d_n$. Every other observation has the correct initial component margin at least $g_0 d_n/4$ after the above score perturbations. After a common label alignment, the first E-step therefore satisfies

$$\frac{1}{n} \sum_{i,a} |R_{ia} - Z_{ia}| \leq \frac{2a_n}{\tau_n} + 2e^{-g_0 d_n/4} = o(1).$$

The graph event has failure $O(n^{-H})$, so these responsibilities meet the reliability condition in Theorem 3.6. The evaluated fitting loop always performs its first M-step before it may stop. That theorem gives the claimed exponent for the first resulting decode and all subsequent finite updates, including the evaluated stopping rule. Its exact-recovery implication also applies. $\square$

Proof of Theorem 3.9. Again choose a fixed $H > 2 + \limsup_{n \rightarrow \infty} I_n / \log n$ and condition on the event in Lemma 3.5. Let $a_n \rightarrow 0$ bound its average profile error, and write $c_0 d_n / n \leq q_{ij} \leq C_H d_n / n$. True profiles have the same positive box after enlarging C_H . Full rank implies distinct block rows; their limiting positive proportions give constants $\kappa, s > 0$ such that

$$\min_{a \neq b} D_P(p_{a*} \| p_{b*}) \geq \kappa d_n, \quad \min_{a \neq b} \|p_{a*} - p_{b*}\|_1 \geq s d_n$$

for all sufficiently large n .

Reliable seed coverage. On this box the two derivatives of the scalar deviance are $\log(x/y)$ and $1 - x/y$, which are uniformly bounded. Consequently

$$0 \leq D_P(x \| y) \leq C d_n, \\ |D_P(x \| y) - D_P(x' \| y')| \leq C(\|x - x'\|_1 + \|y - y'\|_1).$$

Set $\tau_n = \sqrt{a_n} + d_n^{-1}$. Call row i good when $\|q_i - p_{z_i,*}\|_1 \leq \tau_n d_n$; the remaining rows number at most $(a_n / \tau_n)n$. Conditional on previously selected seeds being good and representing distinct classes, every good row in a represented class has residual at most $2C\tau_n d_n$. Every good row in an unrepresented class has residual at least $\kappa d_n / 2$, by the Lipschitz bound. Before K seeds have been selected, an unrepresented class supplies at least $\pi_{\min} n / 2$ good rows. Hence total sampling mass is at least $c n d_n$, whereas the mass of represented good rows and all bad rows is at most

$$C(\tau_n + a_n / \tau_n) n d_n.$$

The first uniform seed is bad with probability at most a_n / τ_n . A union bound over the fixed K choices gives single-run failure at most

$$r_{H,n} \leq C_K(\tau_n + a_n / \tau_n) = o(1).$$

The all-zero fallback cannot occur on this successful path while a class remains unrepresented. Conditional on q , the restart streams are independent. Their joint failure probability is at most

$$r_{H,n}^{[\log n]} = n^{-\omega(1)}.$$

This amplification requires no knowledge of H or the separation constants.

Likelihood selection certifies the winning fit. Define the observation-only saturation constant and soft deviance

$$S_q = \sum_{i,j} (q_{ij} \log q_{ij} - q_{ij}), \quad F(\hat{\pi}, \hat{p}) = - \sum_i \log \sum_k \hat{\pi}_k e^{-D_P(q_i \| \hat{p}_{k*})}.$$

There is the exact identity

$$\mathcal{L} = S_q - F, \quad F \geq \sum_i \min_k D_P(q_i \| \hat{p}_{k*}) \geq 0.$$

Observation Gamma terms, if restored, are common to every fit. A successful run starts with one good seed for each class and uniform weights. Thus

$$F_{\text{initial}} \leq C(a_n + \tau_n)nd_n + n \log K =: \epsilon_n nd_n, \quad \epsilon_n \rightarrow 0.$$

Every exact EM update increases $\mathcal{L}$ and decreases F . Selecting the largest final $\mathcal{L}$ across all runs, including the growing endpoint, therefore gives

$$F_{\text{selected}} \leq \epsilon_n nd_n.$$

This argument does not require identifying the successful run, fitting any run to stationarity, or proving anything about the growing branch's basin dynamics.

All candidate means remain in the same box, since they start at rows or the global mean of q and each M-step forms a convex combination. Keeping the previous mean in a zero-mass component preserves the box too. Strong convexity of $x \log x - x$ on the box, followed by Cauchy–Schwarz, gives

$$D_P(x||y) \geq \frac{n\|x - y\|_2^2}{2C_H d_n} \geq \frac{\|x - y\|_1^2}{2C_H d_n}.$$

Consequently

$$\sum_i \min_k \|q_i - \hat{p}_{k*}\|_1^2 \leq 2C_H \epsilon_n nd_n^2.$$

Put $\eta_n = \sqrt{a_n} + \epsilon_n^{1/4} + d_n^{-1}$. The fractions of rows farther than $\eta_n d_n$ from their true profile, or from every fitted profile, are at most a_n/η_n and $2C_H \epsilon_n/\eta_n^2$, respectively; both tend to zero. Each positive-proportion true class has a row avoiding both bad sets. Its fitted center is within $2\eta_n d_n$ of its true center. The separation sd_n forces different classes to use different fitted centers; with exactly K centers this gives a single label permutation satisfying

$$\max_a \|\hat{p}_{a*} - p_{a*}\|_1 \leq 2\eta_n d_n.$$

The selected responsibilities are weakly consistent. For a row with $\|q_i - p_{z_i,*}\|_1 \leq \eta_n d_n$, every wrongly matched center is at ℓ_1 distance at least $sd_n/2$ for large n , hence its deviance is at least $c_* d_n$, with $c_* = s^2/(8C_H) > 0$. For the selected fit's exact E-step, write

$$R_{ik} = \frac{\hat{\pi}_k e^{-D_P(q_i||\hat{p}_{k*})}}{\sum_l \hat{\pi}_l e^{-D_P(q_i||\hat{p}_{l*})}}.$$

The variational identity is

$$-\log \sum_k \hat{\pi}_k e^{-D_{ik}} = \sum_k R_{ik} D_{ik} + \text{KL}(R_i||\hat{\pi}) \geq \sum_k R_{ik} D_{ik}.$$

Summing over the typical rows and bounding each remaining row's responsibility error by two gives

$$\frac{1}{n} \sum_{i,k} |R_{ik} - Z_{ik}| \leq \frac{2a_n}{\eta_n} + \frac{2\epsilon_n}{c_*} =: \gamma_n \rightarrow 0$$

after the above permutation. The identity remains valid with zero weights using the usual zero-mass conventions, and also with a feasible fixed weight floor.

This weak-start conclusion fails only if the graph event fails or every restart fails, a probability at most $O(n^{-H}) + n^{-\omega(1)}$. Apply Theorem 3.6 to the mandatory final M-step. Since $d_n = \log n$ and $H > 2 + \limsup_{n \rightarrow \infty} I_n/\log n$, the initialization failures are negligible at the target exponent scale. This proves the complete decoder's expected-error bound and its exact-recovery consequence. Prescribed polynomial EM iteration limits and $\lceil \log n \rceil$ restarts give polynomial computational cost. $\square$

A.7 Row-independent projection errors

Proof of Theorem 3.10. Condition on the data generating Π_i and on H_i . Work in the $n - 1$ coordinates other than i , and write $\Pi_* = \Pi_{*,i}$, $\Pi = \Pi_i$, $w = w_{-i}^{ab}$, $p_j = \mathbb{E}A_{ij}$, and $e = (\Pi - \text{Id})w$. The fixed block assumptions imply

$$w \in \text{col}(\Pi_*), \quad p \in \text{col}(\Pi_*), \quad \|w\|_2 \leq C\sqrt{n}, \quad \|w\|_\infty \leq C, \quad \|p\|_2 \leq Cd_n/\sqrt{n}.$$

The constants are uniform over the finitely many contrasts and true labels. For orthogonal projectors,

$$\|\Pi_*(\text{Id} - \Pi)\Pi_*\|_{\text{op}} = \|(\text{Id} - \Pi)\Pi_*\|_{\text{op}}^2 \leq \eta_n^2,$$

because the left matrix is $[(\text{Id} - \Pi)\Pi_*]^\top[(\text{Id} - \Pi)\Pi_*]$. It follows that the mean projection error has the quadratic bound

$$|p^\top e| \leq \|p\|_2 \|w\|_2 \eta_n^2 \leq Cd_n \eta_n^2. \quad (\text{A.6})$$

Also, $\|e\|_2 \leq C\sqrt{n}\eta_n$. By Cauchy–Schwarz and projector leverage,

$$|(\Pi w)_j| \leq \|\Pi e_j\|_2 \|w\|_2 = \sqrt{\Pi_{jj}} \|w\|_2 \leq C\sqrt{L},$$

so $\|e\|_\infty \leq M$ for a fixed positive constant M .

Conditional independence makes $X_j = (A_{ij} - p_j)e_j$ independent and centered. They satisfy $|X_j| \leq M$ and

$$V = \sum_j \mathbb{E}X_j^2 \leq b_+ d_n/n \|e\|_2^2 \leq Cd_n \eta_n^2. \quad (\text{A.7})$$

Here a Bennett bound is stronger than the simpler Bernstein bound at the required scale. We include its short derivation. For a centered X with $|X| \leq M$ and $\lambda \geq 0$, Taylor expansion and $|X|^k \leq M^{k-2}X^2$ imply

$$\mathbb{E}e^{\lambda X} \leq 1 + \frac{\mathbb{E}X^2}{M^2}(e^{\lambda M} - 1 - \lambda M).$$

After multiplying moment-generating functions and using $\log(1+x) \leq x$, exponential Markov inequality optimized in λ gives

$$\Pr\{|\sum_j X_j| > t \mid \Pi\} \leq 2 \exp\left\{-\frac{V}{M^2} h\left(\frac{Mt}{V}\right)\right\}, \quad h(u) = (1+u)\log(1+u) - u, \quad (\text{A.8})$$

with the evident interpretation when $V = 0$.

To choose a uniform vanishing error scale, put $a_n = \max\{\eta_n^2, e^{-d_n}\}$, $s_n = \log(1/a_n)$. Both $a_n \rightarrow 0$ and $s_n \rightarrow \infty$. Take $t = d_n/\sqrt{s_n}$. The bound (A.8) remains valid with V replaced by the upper bound $Cd_n a_n$, since its exponent is nonincreasing in the variance argument. As $u \rightarrow \infty$, $h(u) \sim u \log u$, so

$$\frac{Cd_n a_n}{M^2} h\left(\frac{M}{Ca_n \sqrt{s_n}}\right) = \frac{d_n}{M \sqrt{s_n}} \{s_n - \frac{1}{2} \log s_n + O(1)\} (1 + o(1)) \geq cd_n \sqrt{s_n}$$

for all sufficiently large n . Combining this with (A.6) gives, for $\tau_n = d_n/\sqrt{s_n} + Cd_n \eta_n^2 = o(d_n)$,

$$\Pr\{|A_{i,-i}(\Pi - \text{Id})w| > \tau_n \mid \Pi, H_i\} \leq 2e^{-cd_n \sqrt{s_n}}.$$

Union bound over the finitely many contrasts and addition of $\Pr(H_i^c)$ prove (3.22) with $h_n = c\sqrt{s_n} \rightarrow \infty$. The last claim is an application of Theorem 3.4. $\square$

B Proofs of initialization and optimization properties

B.1 Frozen-candidate greedy gains

Proposition B.1 (Greedy gain guarantee). *Let s_{ic} be fixed finite candidate scores and let S_0 be a fixed nonempty seed set. For candidate sets T , define*

$$G(T) = \sum_i \max_{c \in S_0 \cup T} s_{ic} - \sum_i \max_{c \in S_0} s_{ic}.$$

Then G is normalized, monotone, and submodular. If $m \geq 1$ greedy steps select a set T_m and T^ maximizes G over sets of at most m candidates, then*

$$G(T_m) \geq \{1 - (1 - 1/m)^m\} G(T^*) \geq (1 - e^{-1}) G(T^*).$$

Proof. Normalization and monotonicity follow immediately from the definition. The marginal gain of candidate c is

$$G(T \cup \{c\}) - G(T) = \sum_i \left(s_{ic} - \max_{b \in S_0 \cup T} s_{ib} \right)_+.$$

If $T \subseteq T'$, each maximum in the right side increases, so the marginal gain decreases. This proves submodularity. For the greedy set T_t at time t , monotonicity and telescoping diminishing returns give

$$\begin{aligned} G(T^*) - G(T_t) &\leq G(T_t \cup T^*) - G(T_t) \\ &\leq \sum_{c \in T^* \setminus T_t} \{G(T_t \cup \{c\}) - G(T_t)\} \\ &\leq m \{G(T_{t+1}) - G(T_t)\}. \end{aligned}$$

Starting from $G(T_0) = 0$ and iterating this inequality proves the claimed factor. The final inequality follows from $(1 - 1/m)^m \leq e^{-1}$, including the case $m = 1$. $\square$

This is the standard cardinality-constrained submodular greedy guarantee [8]. It controls added explanation relative to the fixed initial seed, not the multiplicative approximation of a possibly negative total log score. It does not apply unchanged when intermediate EM steps alter the candidate scores.

B.2 Noiseless profile selection

Proposition B.2 (Population profile selection). *Suppose $q_i = \theta_{z_i}$, where the K positive profile vectors $\theta_1, \dots, \theta_K$ are pairwise distinct and every class is nonempty. The candidate pool consists of these node profiles. From any first seed belonging to this candidate pool, $K - 1$ subsequent global-gain greedy steps select all K different profiles. Maximum working-score classification using those profiles recovers z up to permutation.*

Proof. For $x, \hat{p} > 0$ coordinatewise, write $\ell_x(\hat{p}) = \sum_j x_j \log \hat{p}_j - \sum_j \hat{p}_j$. Then

$$\ell_x(x) - \ell_x(\hat{p}) = \sum_j \{x_j \log(x_j / \hat{p}_j) - x_j + \hat{p}_j\} = D_P(x \| \hat{p}) \geq 0,$$

with equality precisely when $x = \hat{p}$; the inequality follows from $\log u \leq u - 1$. If a profile has already been selected, selecting a duplicate has zero gain. If profile θ_b has not been selected, then each node in class b gains $\min_{a \in S} D_P(\theta_b \| \theta_a) > 0$ when it is added. Therefore some unselected profile has strictly positive gain, and greedy cannot select a duplicate before all distinct profiles have been selected. The same strict divergence inequality makes each node's own profile its unique best score after selection. $\square$

The proposition concerns the frozen node-profile selection control in an abstract noiseless profile model. It does not directly cover Algorithm 1, whose initial mean need not be a node profile and whose intermediate EM fits change the parameters. It does not replace a stability analysis for random spectral profiles, and it does not erase the diagonal distinction in a no-self-loop SBM.

B.3 Exact working EM and accepted replacements

Proposition B.3 (Objective convergence of exact working EM). *For fixed strictly positive profiles $q_1, \dots, q_n$, let*

$$\mathcal{L}(\hat{\pi}, \hat{p}) = \sum_i \log \sum_{a=1}^K \hat{\pi}_a \exp\{\ell_i(\hat{p}_a)\}, \quad \ell_i(\hat{p}_a) = \sum_j q_{ij} \log \hat{p}_{aj} - \sum_j \hat{p}_{aj}.$$

Starting with positive parameters and positive mixture weights, the exact E-step and unconstrained exact M-step

$$r_{ia} = \frac{\hat{\pi}_a e^{\ell_i(\hat{p}_a)}}{\sum_b \hat{\pi}_b e^{\ell_i(\hat{p}_b)}}, \quad N_a = \sum_i r_{ia},$$

$$\hat{\pi}_a^{\text{new}} = N_a/n, \quad \hat{p}_{aj}^{\text{new}} = \frac{\sum_i r_{ia} q_{ij}}{N_a}$$

make $\mathcal{L}$ nondecreasing. Its numerical values converge to a finite limit. The same monotonicity statement holds for an exact M-step maximizing over any stated constraint set containing the current parameters, and for replacement proposals accepted only when their final $\mathcal{L}$ improves.

Proof. For row-wise probability vectors r_i , define the variational objective

$$\mathcal{F}(r, \hat{\pi}, \hat{p}) = \sum_{i,a} r_{ia} \{\log \hat{\pi}_a + \ell_i(\hat{p}_a) - \log r_{ia}\}.$$

The log-sum inequality gives $\mathcal{F}(r, \hat{\pi}, \hat{p}) \leq \mathcal{L}(\hat{\pi}, \hat{p})$, with equality for the E-step responsibilities. For fixed r , maximizing $\sum_a N_a \log \hat{\pi}_a$ over the simplex gives $\hat{\pi}_a = N_a/n$. For each profile coordinate, maximizing $(\sum_i r_{ia} q_{ij}) \log \hat{p}_{aj} - N_a \hat{p}_{aj}$ gives the displayed weighted mean. These are interior maxima because all inputs and responsibilities are positive. Thus the E-step makes the variational bound tight and the M-step raises it, proving monotonicity.

The saturated-score inequality used in the preceding proof implies $\ell_i(\hat{p}_a) \leq \ell_i(q_i)$ for every i, a . Since $\sum_a \hat{\pi}_a = 1$,

$$\log \sum_a \hat{\pi}_a e^{\ell_i(\hat{p}_a)} \leq \ell_i(q_i).$$

Consequently $\mathcal{L} \leq \sum_i \ell_i(q_i) < \infty$. A nondecreasing real sequence bounded above has a finite limit. For a constrained exact M-step the current parameters remain feasible, so the same variational argument applies. The replacement assertion follows directly from its acceptance rule. $\square$

This proposition proves convergence of objective values. It proves neither convergence to the global maximum nor recovery of the true labels. The implementation uses an exact active-set maximizer on the floored simplex, so its weight constraint is covered by the constrained version of this proposition. Its finite stopping tolerance and floating-point accuracy are assessed separately in Section 5.

Proposition B.4 (Profile hull and spectral dimension). *Every responsibility-weighted profile M -step produces a profile in the convex hull of the rows of q . If the unmodified reconstruction $q = U\Lambda U^\top$ is used and the resulting means are positive, all fitted profiles lie in its row space, of dimension at most K . Entrywise positive clipping can increase this dimension.*

Proof. The coefficients R_{ia}/N_a are nonnegative and sum to one, so the update is a row convex combination. Every such combination also lies in the row space. The unmodified reconstruction has rank at most K . The clipping operation is nonlinear and has no corresponding rank-preservation identity. $\square$

C Implementation and reproducibility

C.1 Experimental entry points and retained outputs

The numerical evidence is archived in the repository

<https://github.com/zhixin0825/spectral-lrt>.

The complete safeguarded implementation is `Work.certified_growing()` in `residual_seed.py`, exposed as `method='global_gain_certified'`. It uses $\lceil \log n \rceil$ independent residual starts and one mandatory final M -step. The archived primary benchmark remains the original growing branch. The principal comparison is in `experiments/20261009_residual_likelihood_seeding/`. Its report, protocol, model-generation script, fitting code, scalar verification, and per-graph summaries define the numerical experiment. The preceding partial-peeling experiment and its diagnosis are in `experiments/20261009_partial_likelihood_peeling/`. The original growing branch corresponds to the recorded `growing_global_gain_EM` arm and the API option `method='global_gain'`. It starts at the global profile mean, adds $K - 1$ node candidates, and refits after every addition. The `global_gain_no_update` arm is a separate frozen-profile control with a top-subset first seed. Identical final error counts do not identify the algorithms.

The manuscript’s two CSV summaries retain batch-specific, combined, and model-specific results. The diagnostic figure uses saved coordinates and fitted labels rather than a new simulation. Full spectral arrays and objective traces are stored in the previously archived numerical records; the source reports describe their restoration. Reconstructing the profiles from (U, Λ) and fitting the working score does not need the original adjacency matrix.

C.2 Historical subset controls

These controls are retained for reproducibility of earlier experiments and are not steps of Algorithm 1. In the frozen global-gain control, the first candidate maximizes the sum of its largest $m_0 = \min\{\lfloor 2n/(3K) \rfloor, n - K\}$ scores over other nodes, excluding itself. In peeling, when M candidates and r components remain, the selected subset contains $\min\{\lfloor 2M/(3K) \rfloor, M - r\}$ other nodes. Only these historical controls require a subset-size rule; the primary growing algorithm uses the global mean and all-node gains throughout.

C.3 Floored-simplex weight updates

For a fixed floor $f \in [0, 1/K)$, the weight M-step solves

$$\max_{\hat{\pi}_a \geq f, \sum_a \hat{\pi}_a = 1} \sum_a N_a \log \hat{\pi}_a. \quad (\text{C.1})$$

Its active-set solution starts with every component free. If the free set is S , assign mass $1 - f(K - |S|)$ proportionally to the N_a , $a \in S$, and fix any weight below f at the floor. Repeat on the remaining free set. The KKT conditions give

$$\hat{\pi}_a = \max\{f, N_a/\lambda\}, \quad \sum_a \hat{\pi}_a = 1 \quad (\text{C.2})$$

for a suitable multiplier λ . The active-set procedure enforces this equation; it is not a naive clipping followed by renormalization. The tested function `_floored_weights` implements this exact constrained update. Since the E-step and mean M-step are unchanged, the variational proof applies on the floored-simplex parameter domain.

C.4 An explicit error decomposition

For fitted components matched to labels a, b , put

$$\hat{w}_j^{ab} = \log(\hat{p}_{aj}/\hat{p}_{bj}), \quad \hat{c}^{ab} = - \sum_j (\hat{p}_{aj} - \hat{p}_{bj}) + \log(\hat{\pi}_a/\hat{\pi}_b). \quad (\text{C.3})$$

If λ^{ab} is the chosen oracle prior log ratio, the empirical comparison has the decomposition

$$\begin{aligned} \hat{T}_i^{ab} - (L_i^{ab} + \lambda^{ab}) &= A_i(\Pi - \text{Id})w^{ab} + (q_i - \tilde{A}_i)w^{ab} \\ &\quad + q_i(\hat{w}^{ab} - w^{ab}) + \hat{c}^{ab} - c_i^{ab} - \lambda^{ab}. \end{aligned} \quad (\text{C.4})$$

The terms correspond to empirical projection, positivity clipping, estimated contrasts, and intercept calibration. The reconstructed diagonal is included in these terms; the oracle excludes it through $A_{ii} = 0$ and the definition of c_i^{ab} . For Poisson rather than Bernoulli oracle contrasts, the sparse correction must also be included. Formula (C.4) is exact and gives concrete quantities for checking the transfer conditions. Theorem 3.6 controls the empirical fitting stage through a comparison with clipped oracle block-count scores. It does not require a separate rare-event $o(d_n)$ bound on the raw clipping term in this decomposition. The original growing branch has an end-to-end proof when $K = 2$. The likelihood-restart Algorithm 1 has an end-to-end proof for general fixed K .

C.5 Verification of the manuscript

The source comprises six numbered main sections, proof appendices, and a bibliography. A build runs PDFLaTeX, BibTeX, and two additional PDFLaTeX passes, checking unresolved citations and references. Build provenance records the source commit and the checksums of the source and compiled PDF. Compilation checks syntax and cross references; mathematical conclusions are exactly those stated with their hypotheses in the text.

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